A New Auxiliary Model Approach For Fermionic Criticality

Insights on Mottness in Strongly Correlated Systems

ABHIRUP MUKHERJEE

Presubmission Open Seminar

IISER KOLKATA

Emergent Phenomena in Quantum Matter group
IISER Kolkata

EPQM

Acknowledgements

Publications

New J. Phys. 25 113011. **AM**, N S Vidhyadhiraja, A Taraphder, S Lal. (2023)

J. Phys. A: Math. Theor. 57 275401. **AM**, S Patra, S Lal. (2024)

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Phys. Rev. B 105, 085119. A Mukherjee, **AM**, N S. Vidhyadhiraja, A Taraphder, S Lal. (2022)

J. Phys.: Condens. Matter 35 315601. S Patra, **AM**, A Mukherjee, N S Vidhyadhiraja, A Taraphder, S Lal. (2023)

Phys. Rev. B 111, L140414. A Kumar N M, S Mukherjee, **AM**, A Punjal, S Purwar, T Setti, S Prabhu S., S Lal, N Kamaraju. (2025)

arXiv:2602.20600. D Debata, **AM**, S Lal. (2026)

Outline of Thesis

Chap 1. Introduction

Chap 2. Methods and Preliminaries

Chap 3. Kondo frustration via charge fluctuations: Mott localisation in $d = \infty$ [**New J. Phys. 25 113011**]

Chap 4. A New Auxiliary Model Mapping For Correlated Lattice Models [**New J. Phys. 28 013503**]

Chap 5. Mott Criticality as the Confinement Transition of a Pseudogap-Mott Metal in $d = 2$ [**New J. Phys. 28 013503**]

Chap 6. Competing tendencies In Heavy Fermion Systems: Kondo screening vs. Mott localisation [**in progress**]

Chap 7. Holographic entanglement renormalisation for fermionic quantum matter [**J. Phys. A: Math. Theor. 57 275401**]

Chap 8. Conclusions

Introduction To Fermionic Criticality and Quantum Materials

What Do Condensed Matter Theorists Do?

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Explain Experiments

Build and solve theoretical models that explain observations

Examples: Quantum Hall effect, superconductivity

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Predict Experiments

Identify interesting theoretical models that predict novel phenomena

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Predict Experiments

Identify interesting theoretical models that predict novel phenomena

Examples: Abrikosov vortex lattice, spin liquids

Classify experiments

Come up with frameworks that explain a large class of phenomena

Examples: Landau theory of phase transition

Quantum Materials

Class of materials that **cannot** be well-approximated by “classical” description

Quantum Materials

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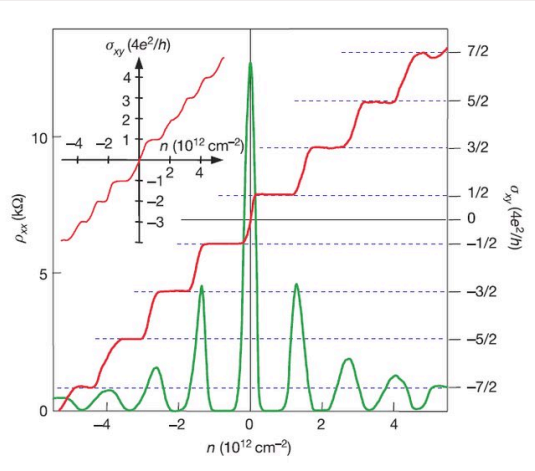

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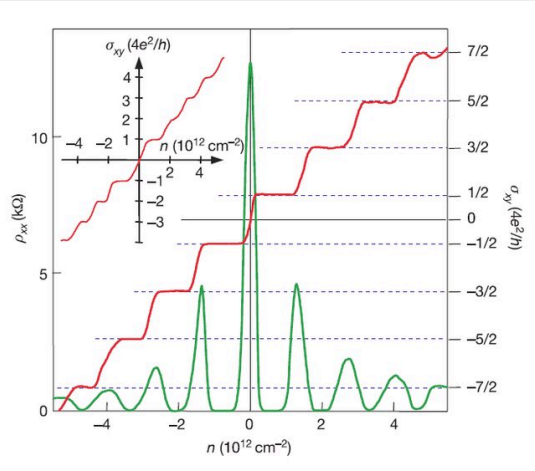

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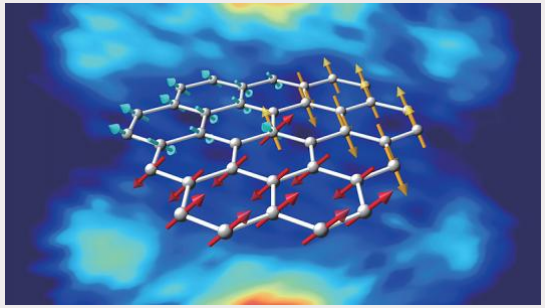

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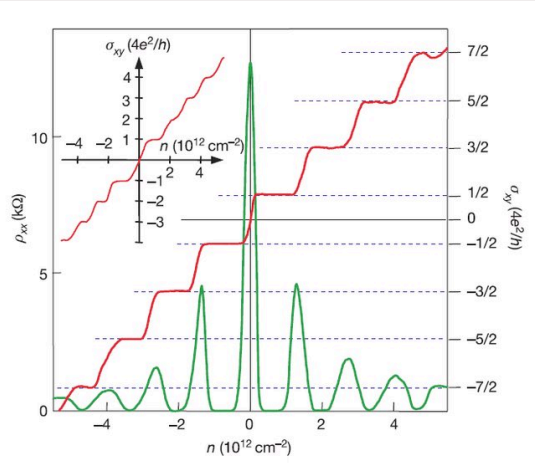

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collective excitations
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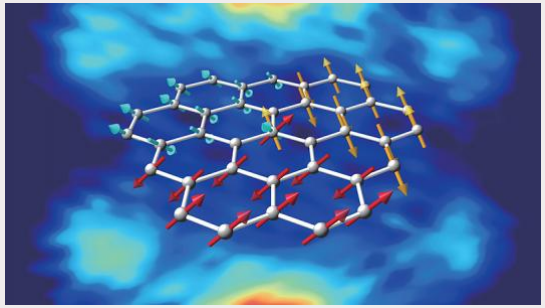

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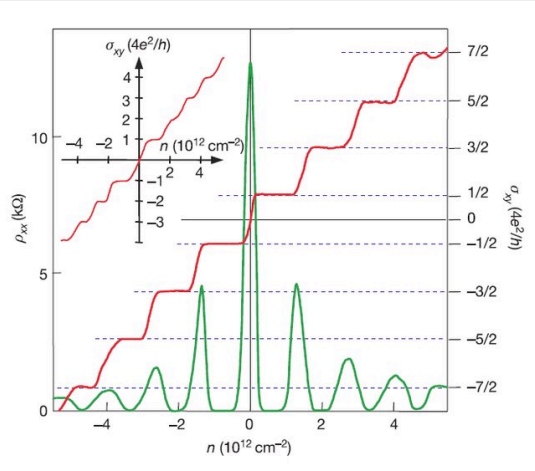

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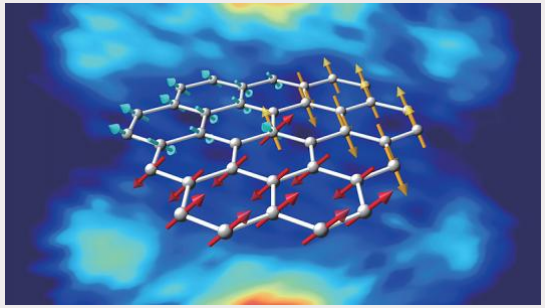

Needs dramatically **new ideas** and methods!

- **non-local/collective** order parameters for transition
- ability to work with a large number of **interacting particles**
- ability to work with **emergent** degrees of freedom

Strongly Correlated Phenomena

Materials with strong **local** correlation, **low dimensionality**, multiple electronic **orbitals**

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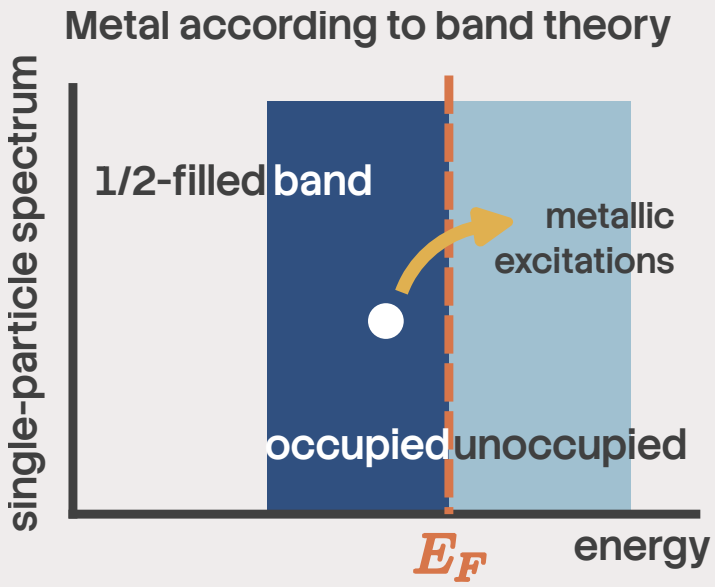
Phenomena: Mott Insulation



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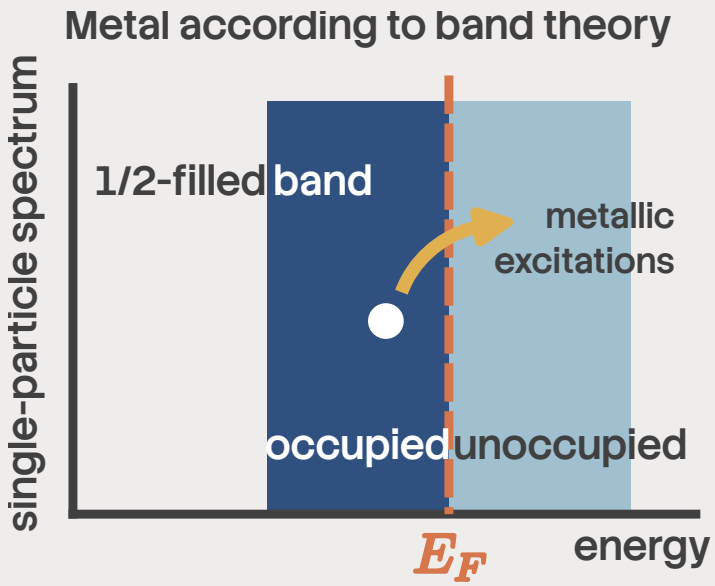

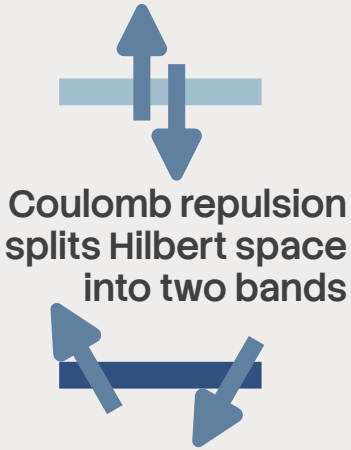

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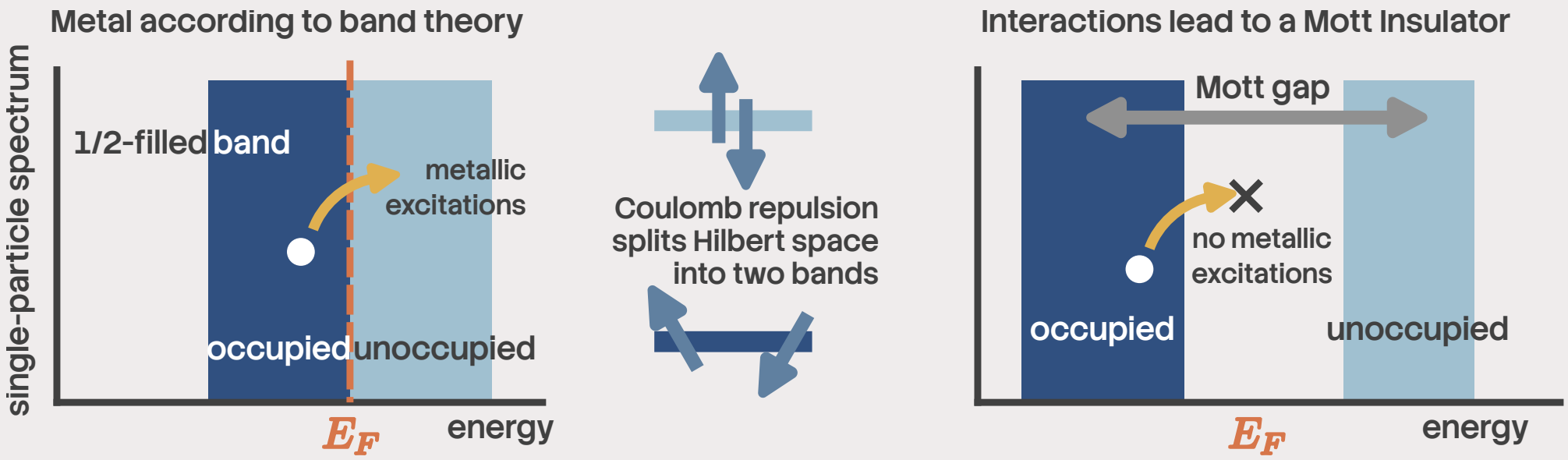

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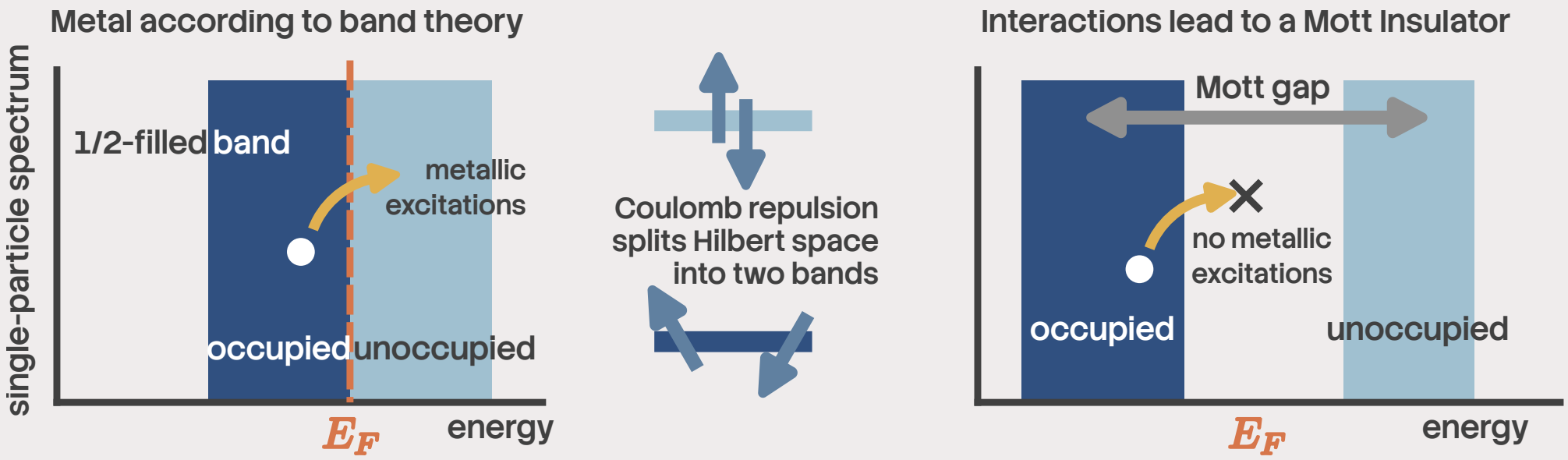

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- result of **manybody effects**

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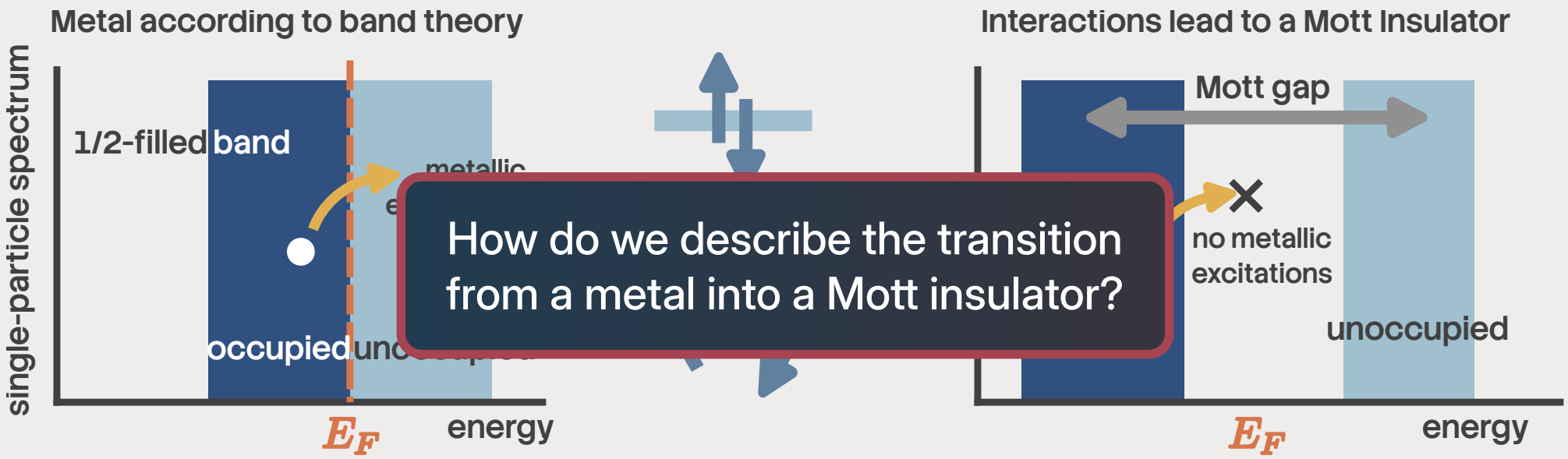

How do we describe the transition from a metal into a Mott insulator?

- Not accessible from 1-particle theory
- result of **manybody effects**

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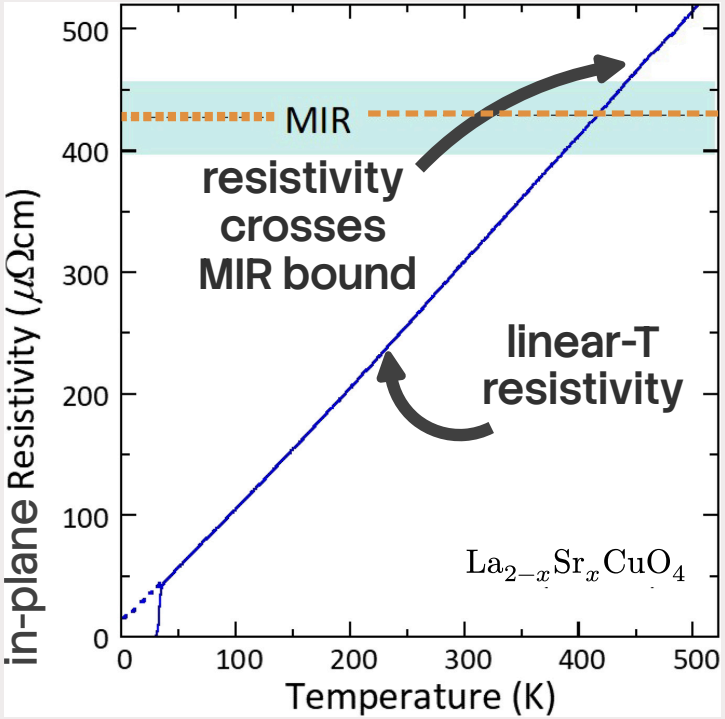

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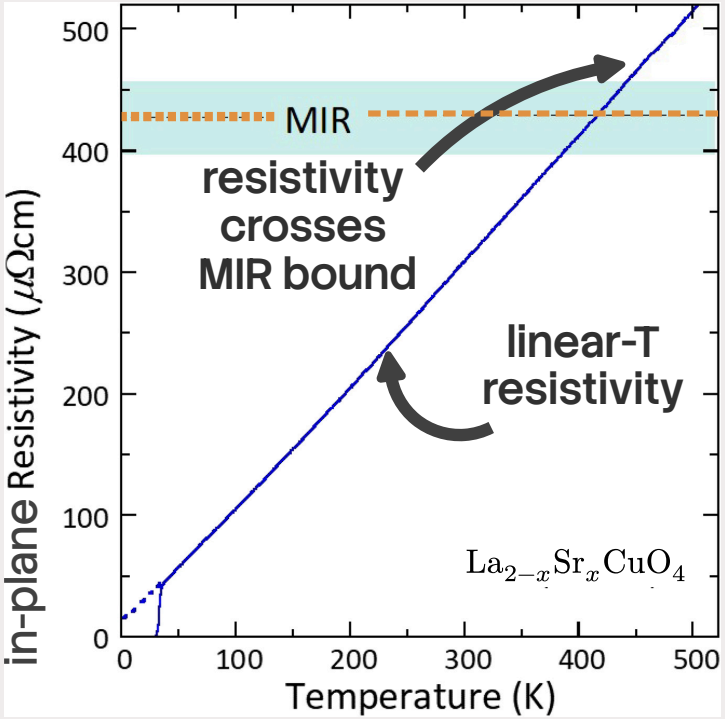
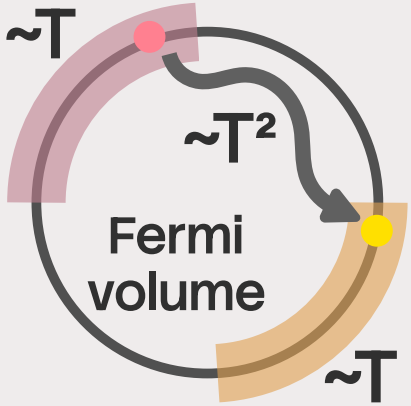


Conventional metallic theory predicts $R \propto T^2$



Source of $R \propto T^2$?

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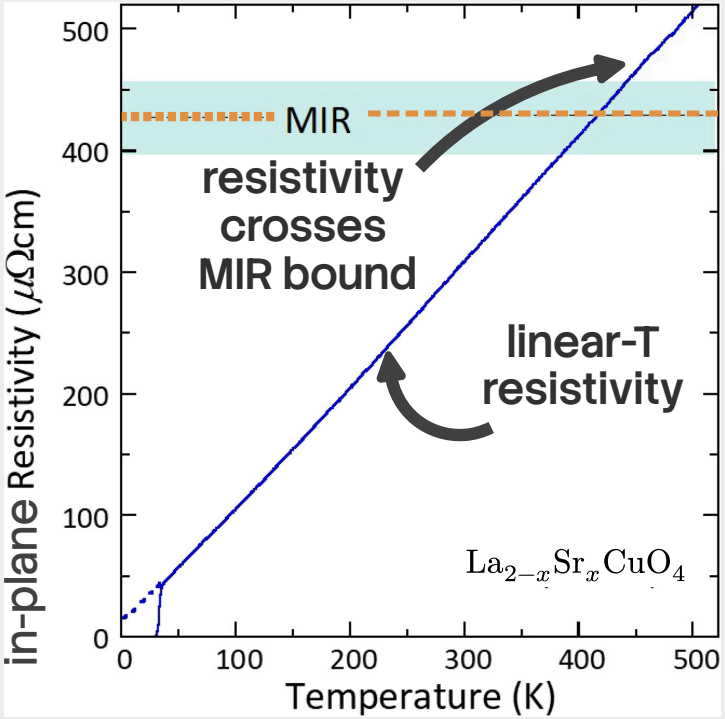

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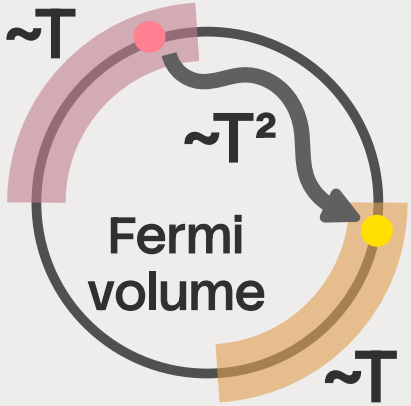

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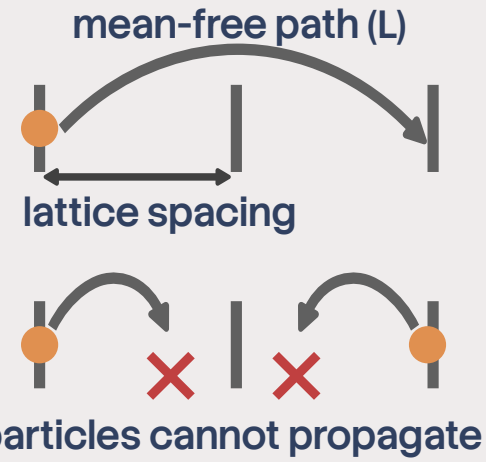
Single particles must propagate finite number of lattice spacing



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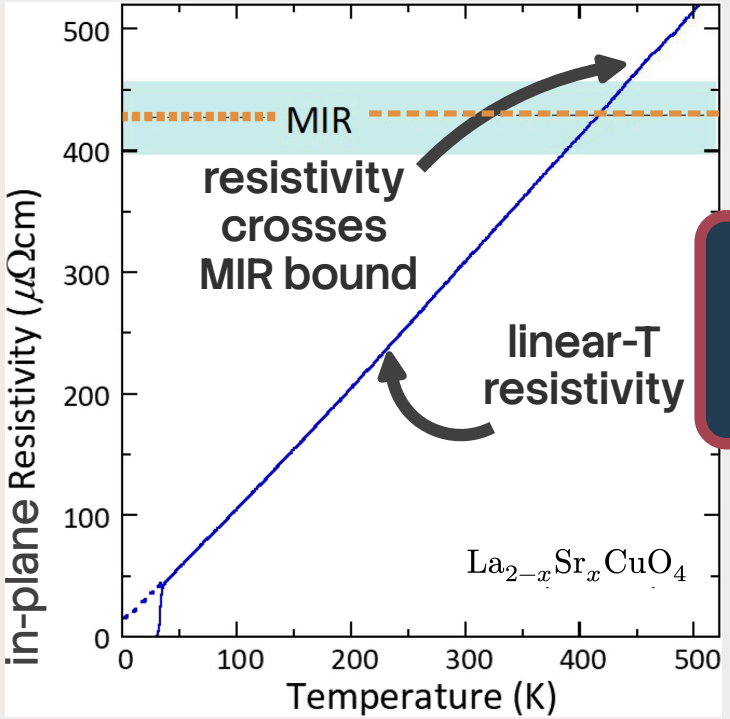

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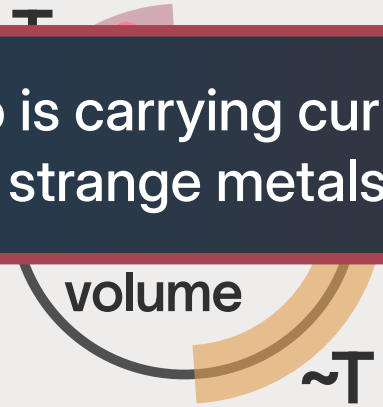

Source of $R \propto T^2$?

Who is carrying current in strange metals?

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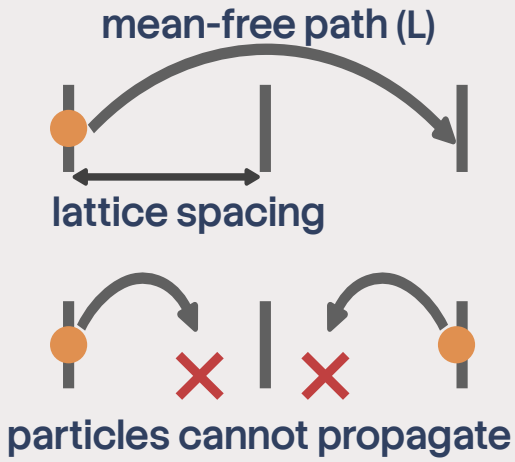

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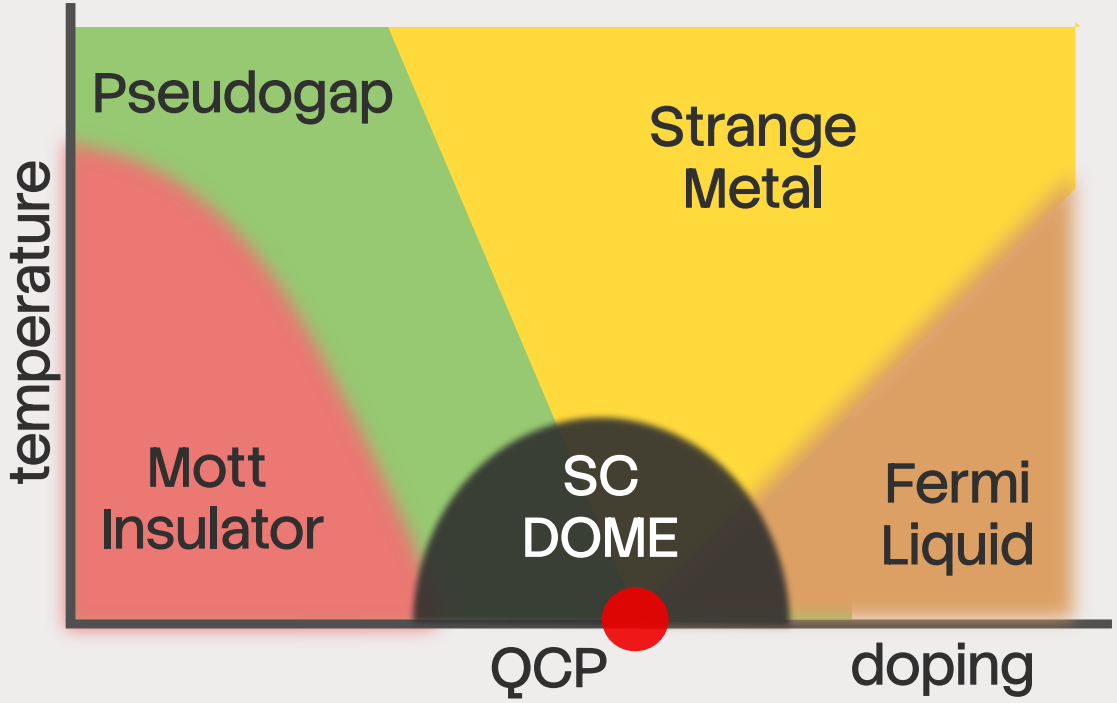
Schematic phase diagram of **copper oxide materials**



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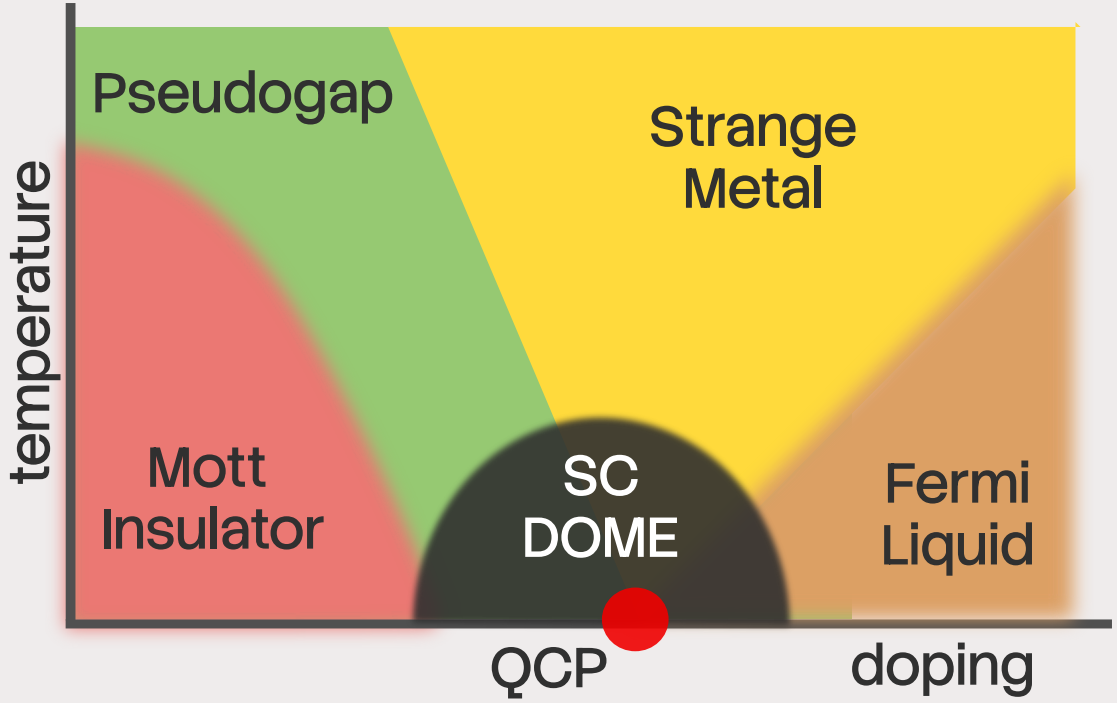

Theoretical Models of Correlated Materials

Hubbard Model

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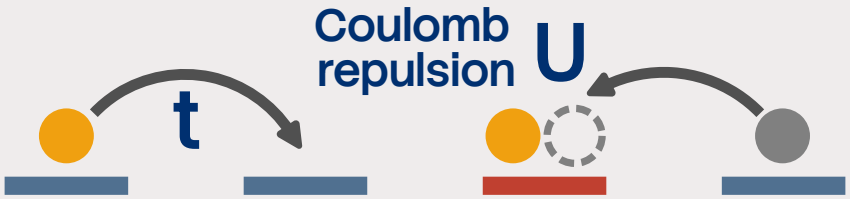
- Energy cost U if two occupy same site.



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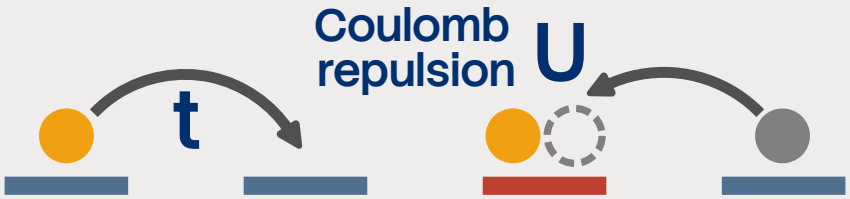

Hamiltonian: $-t \sum_{i,j,\sigma} (c^{\dagger}_{i,\sigma} c_{j,\sigma} + \text{h.c.}) + \sum_i U n_{i,\uparrow} n_{i,\downarrow}$

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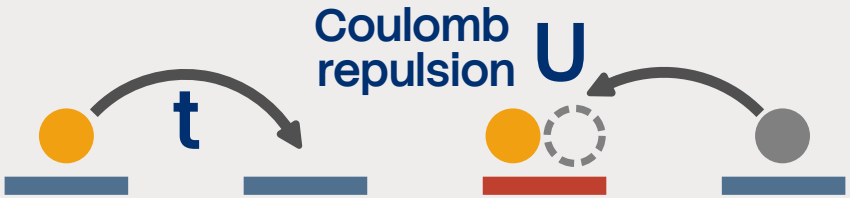

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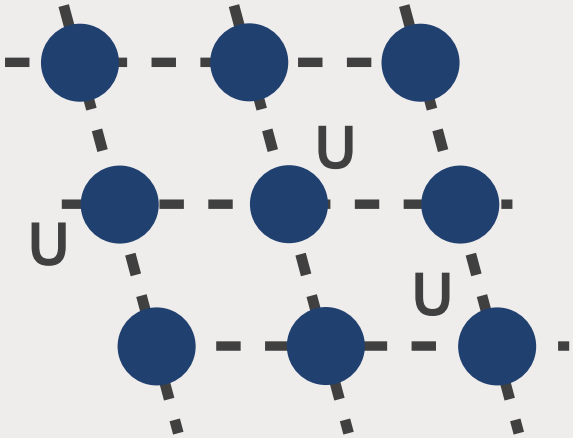
$t \Longrightarrow$ Delocalisation
$U \Longrightarrow$ Localisation

- Energy cost U if two occupy same site.

🔥 Competition
🫂 Correlations

Hamiltonian: $-t \sum_{i,j,\sigma} (c^{\dagger}_{i,\sigma} c_{j,\sigma} + \text{h.c.}) + \sum_i U n_{i,\uparrow} n_{i,\downarrow}$

Quantum Impurity Models as Auxiliary Models



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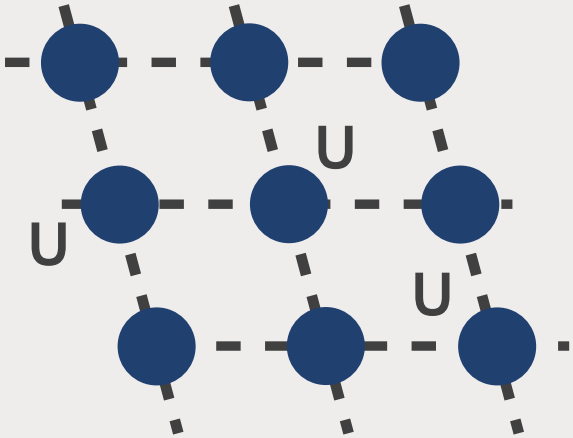

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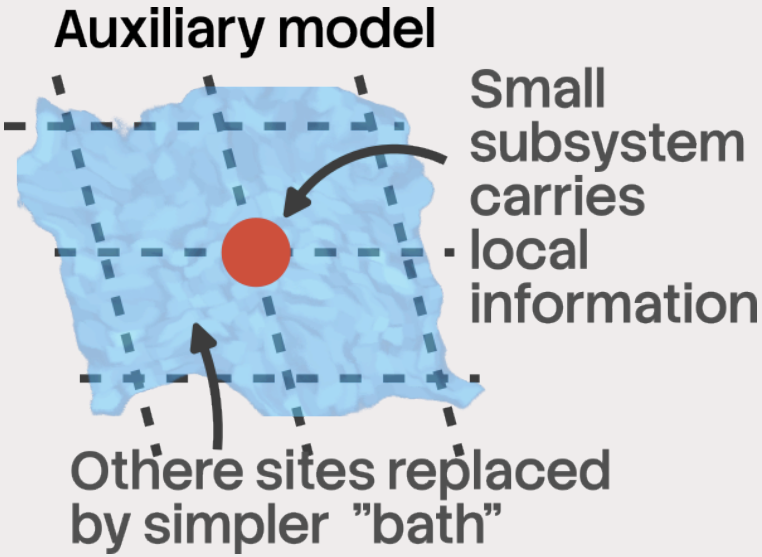
Idea: replace full problem with a simpler **auxiliary model** that captures the local correlations.



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Auxiliary Model for Interacting Quantum Systems?

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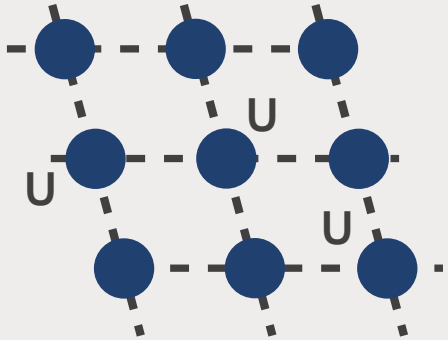
Auxiliary Model has to be **interacting** in order to capture temporal dynamics!

What is an appropriate auxiliary problem for lattice models with local correlation?



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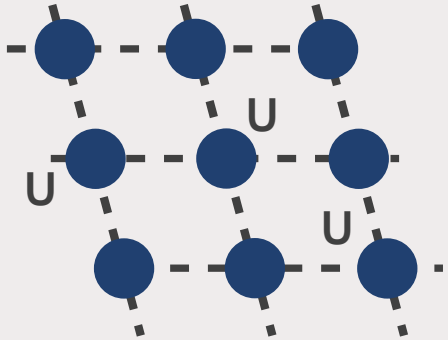

Insight from Limit of Infinite Dimensions!

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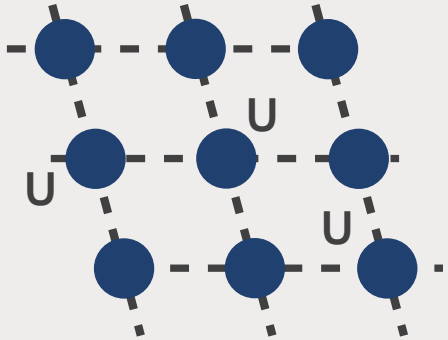

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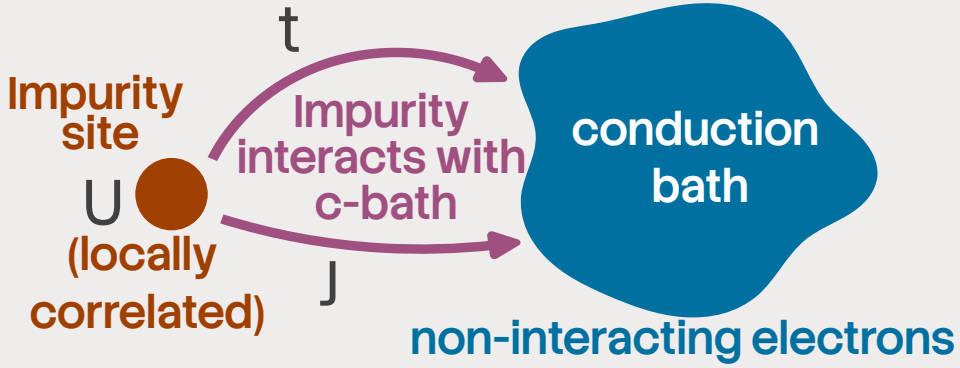
Local physics ⟹ **Quantum Impurity Models**

Core Idea of Dynamical Mean-Field Theory



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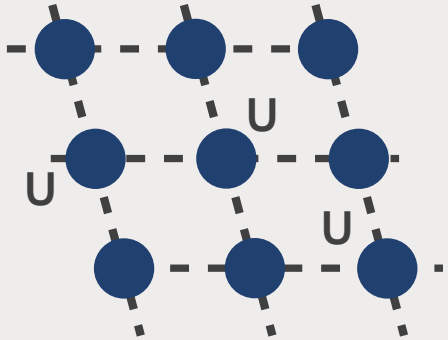

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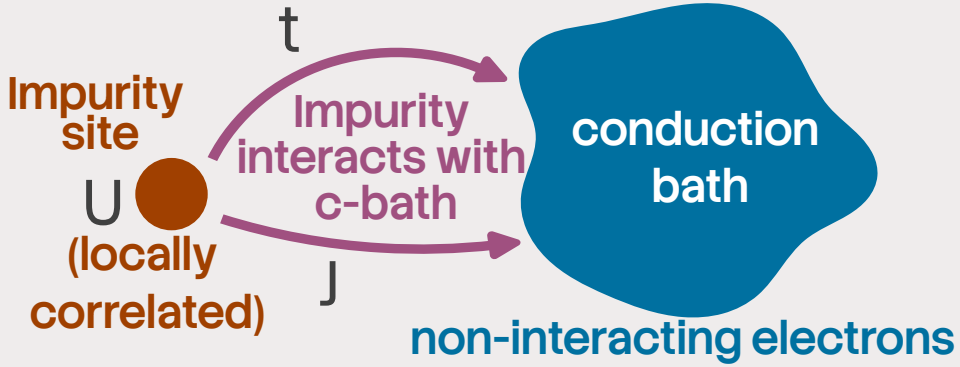

- Tractable
- Contains dynamics

(ω –dependence of correlations)

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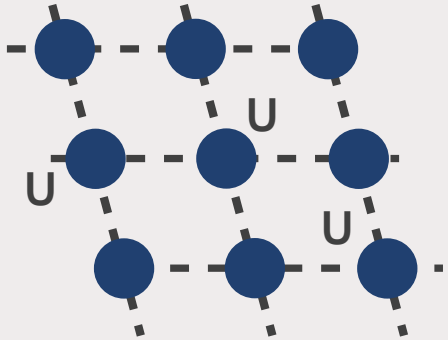
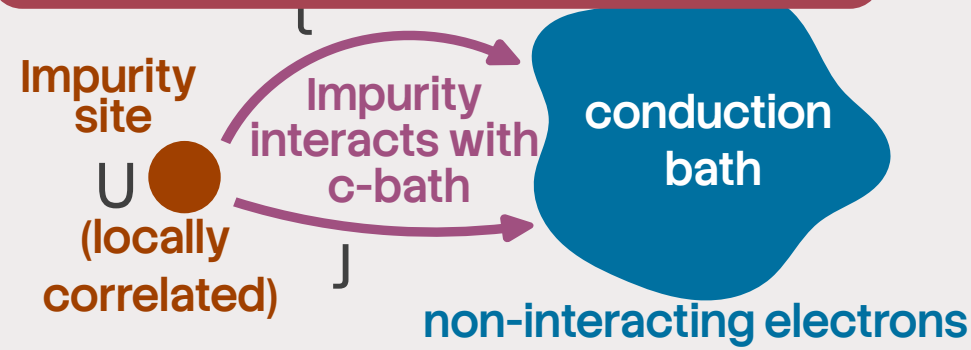


Local physics of correlated lattice models can be mapped on to impurity model.

Insight from Limit of Inf

Local physics ⟹ **Quantum Impurity Models**

Core Idea of Dynamical Mean-Field Theory



- Tractable
- Contains dynamics

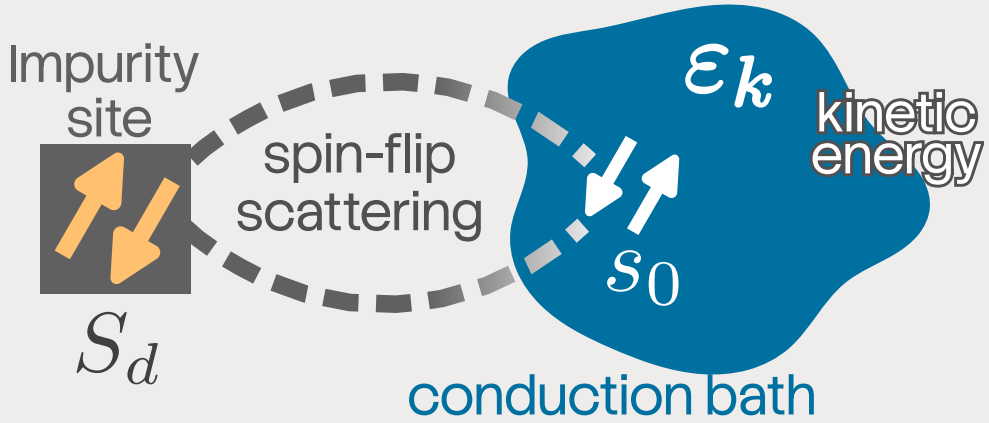
(ω –dependence of correlations)

Phenomenology of Quantum Impurity Models

Kondo Model

- Impurity spin S_d + conduction bath
- Spin-flip scattering

$$H = \sum_{k,\sigma} \varepsilon_k n_{k,\sigma} + \frac{1}{4} J S_d^z s_0^z + \frac{1}{2} S_d^+ s_0^- + \frac{1}{2} S_d^- s_0^+$$



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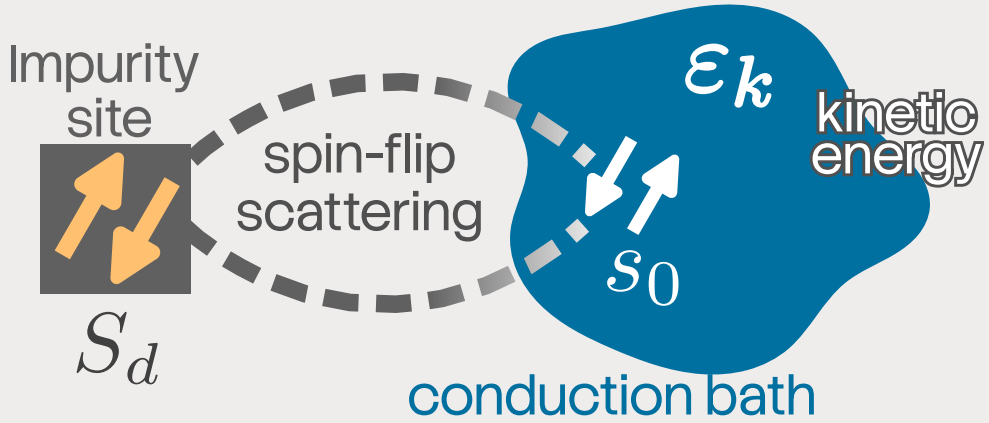
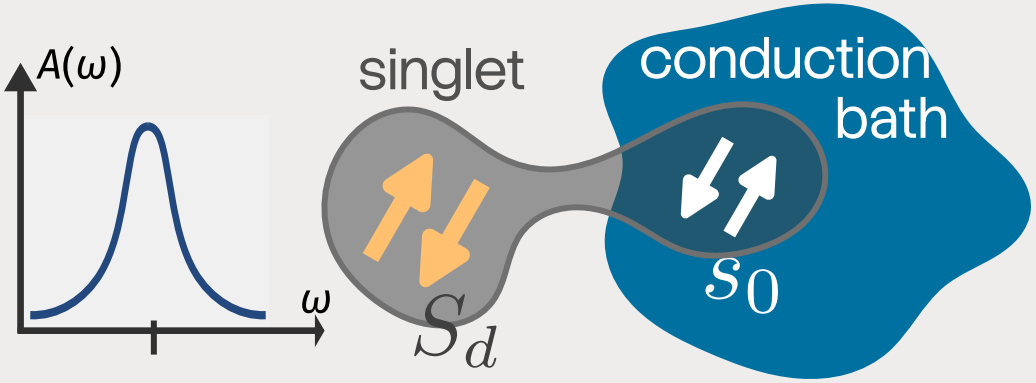


Low-energy ($T \ll 1$) Properties: Kondo Screening

- Kondo coupling generates **entanglement**
- Impurity DOS supports **gapless excitations**
- Impurity electron can **delocalise** via bath



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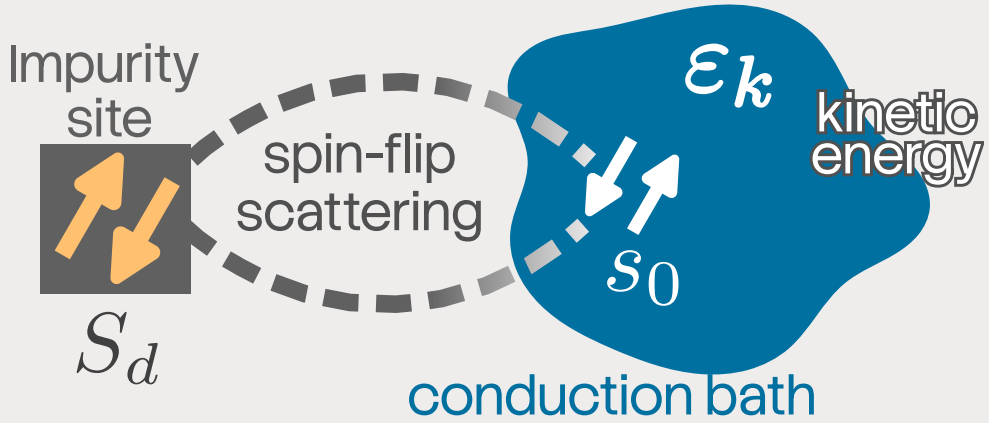

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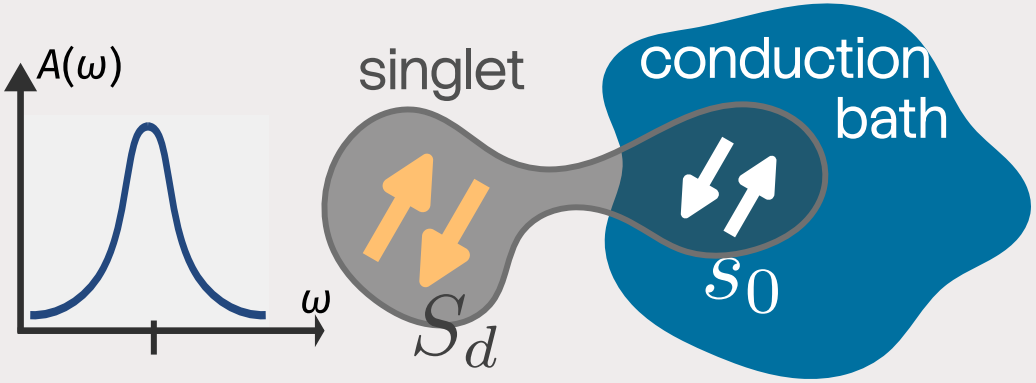
$\Longrightarrow$ **metallic state** in auxiliary model language.



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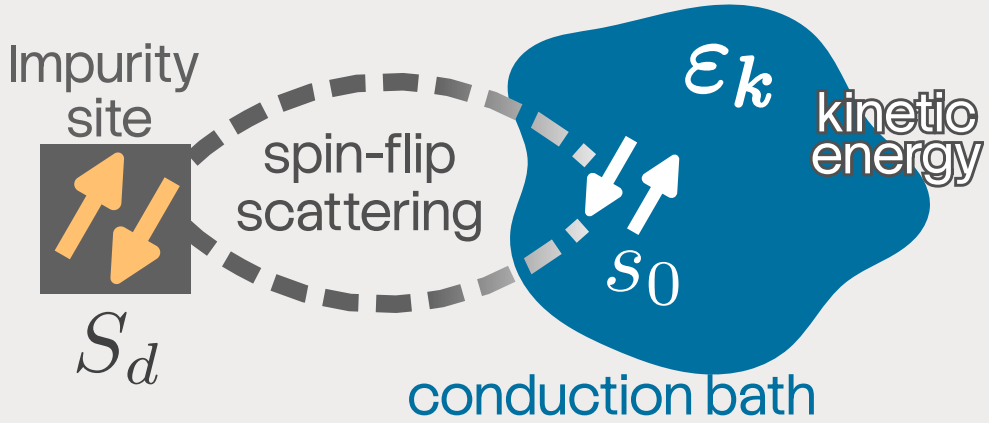
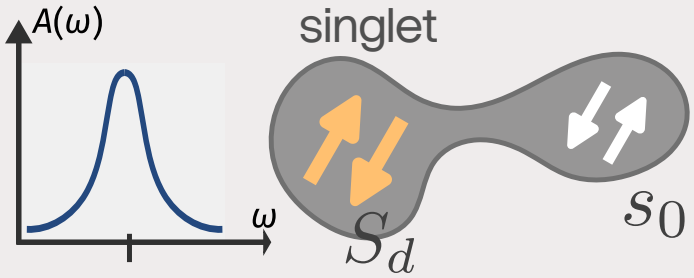


Metallic State



How Do We Get an Insulating State?

Phenomenology of Quantum Impurity Models

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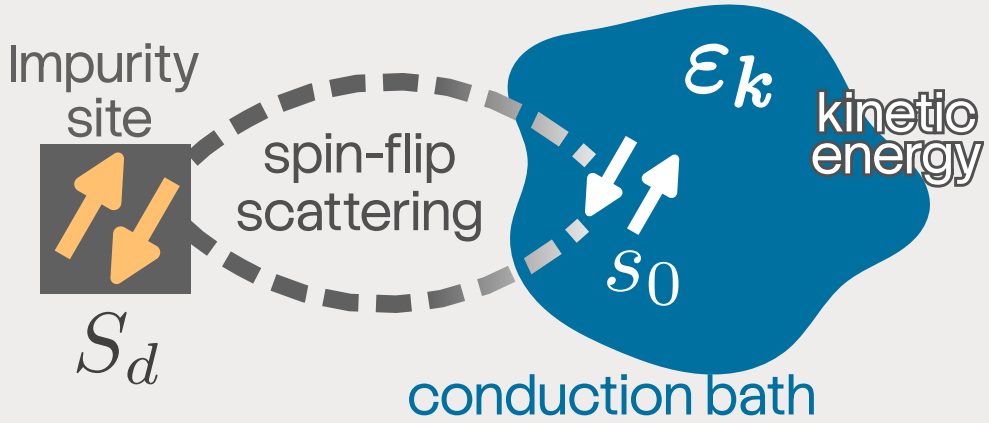

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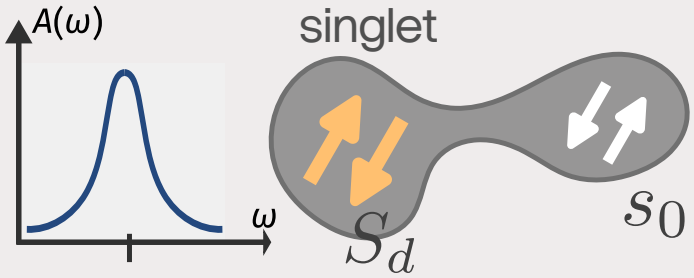
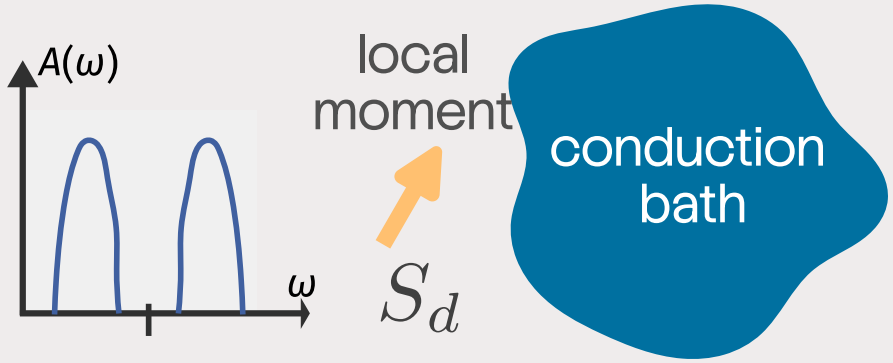


How Do We Get an Insulating State?

- Tweak the bath
- Destroy screening
- Local moment emerges



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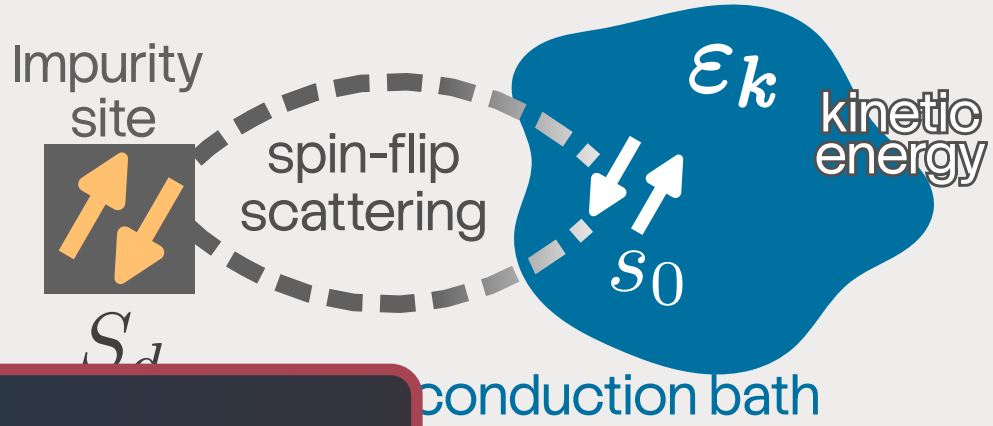
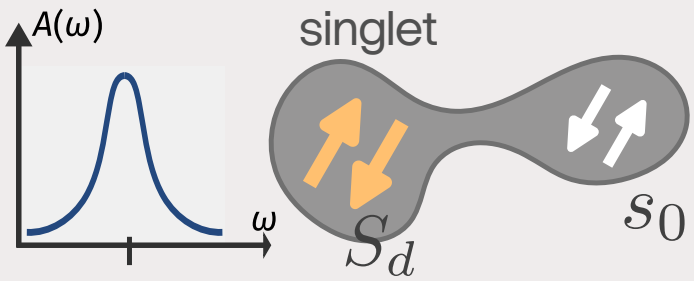


Kondo screening ⟹ Metallicity

Local moment ⟹ Mott insulation

Metallic State



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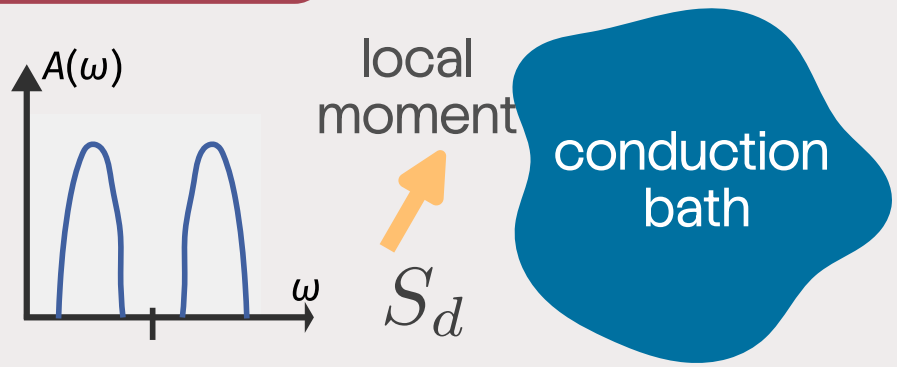
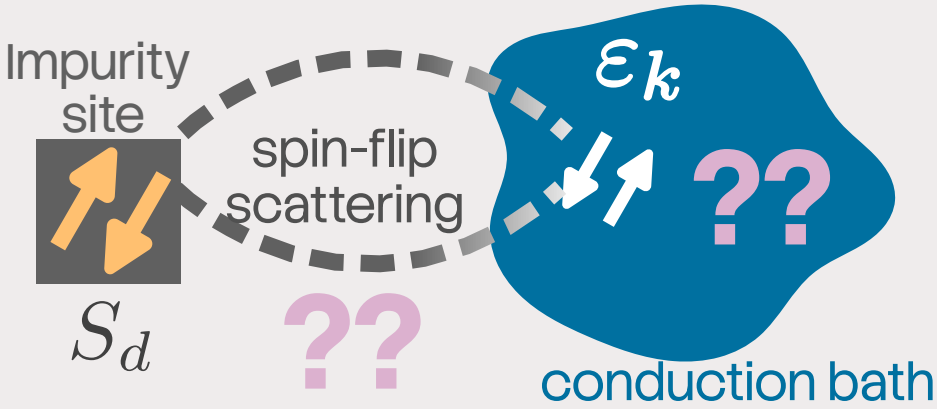
Lating State?



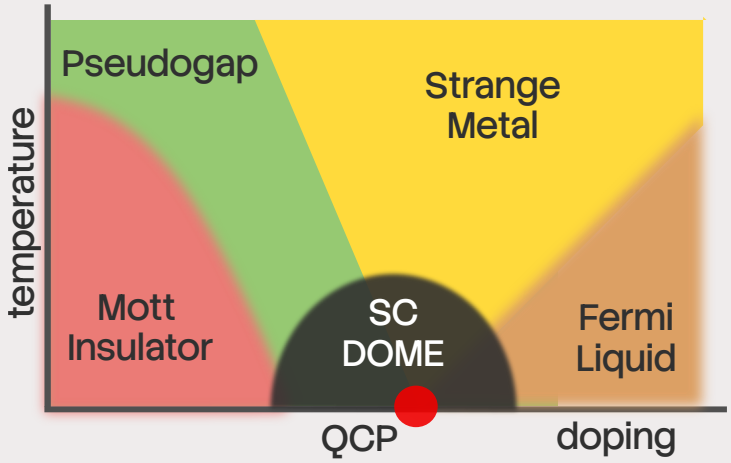
Outstanding Questions

Appropriate impurity model to describe the Mott transition (locally)?



General Approach To Incorporate Non-Local Fluctuations?

Relation Between Various Correlated Phases?



Minimal Model For Kondo Breakdown and Mott Transition

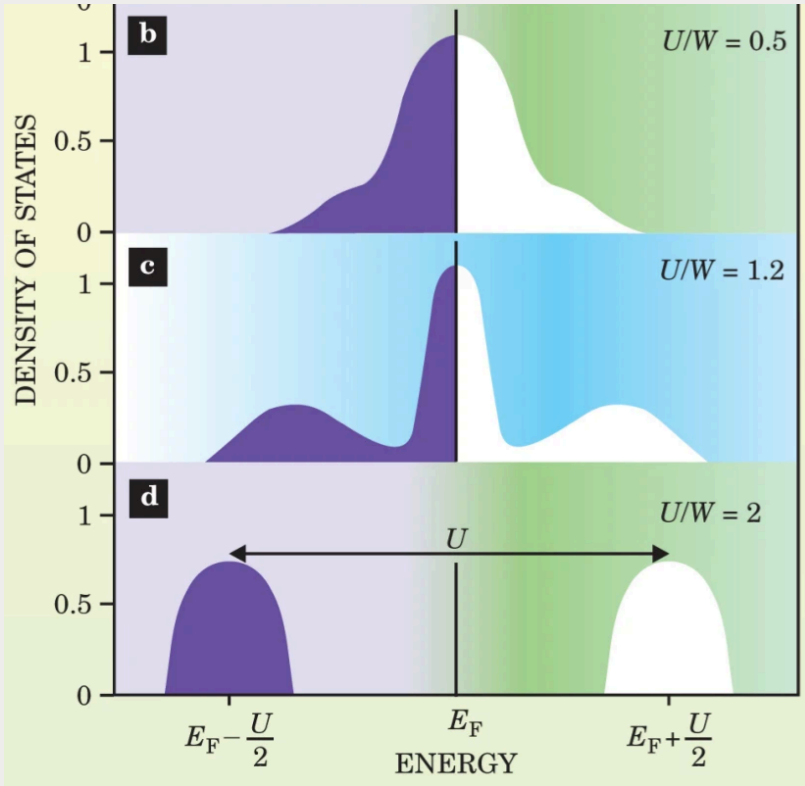
A Route to Mott Localisation in Infinite Dimensions

New J. Phys. 25 113011 (2023)
AM, N S Vidhyadhiraja, A Taraphder, S Lal

Mott Transition In Infinite Dimensions

DMFT Results on Hubbard Model in $D = \infty$

Dynamical mean-field theory sees a metal-insulator transition by **self-consistent** determination of bath.



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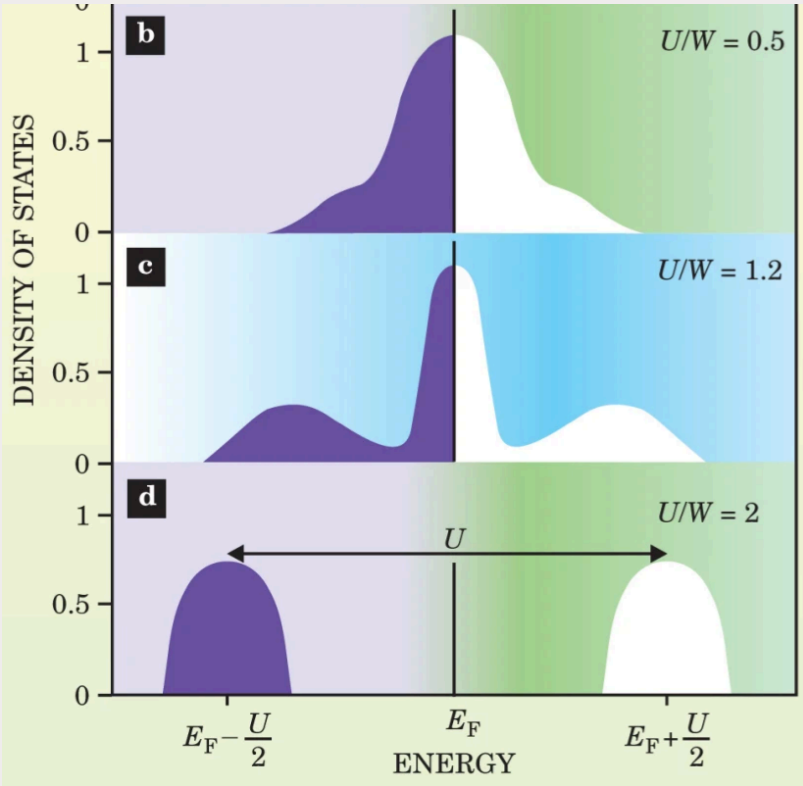

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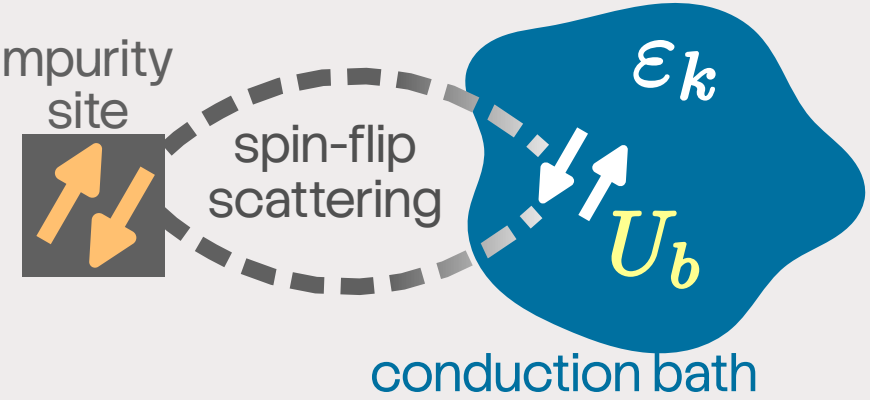
Our Approach: Add Local Correlation in Bath

$-|U_b|\, n_{0\uparrow} n_{0\downarrow}$



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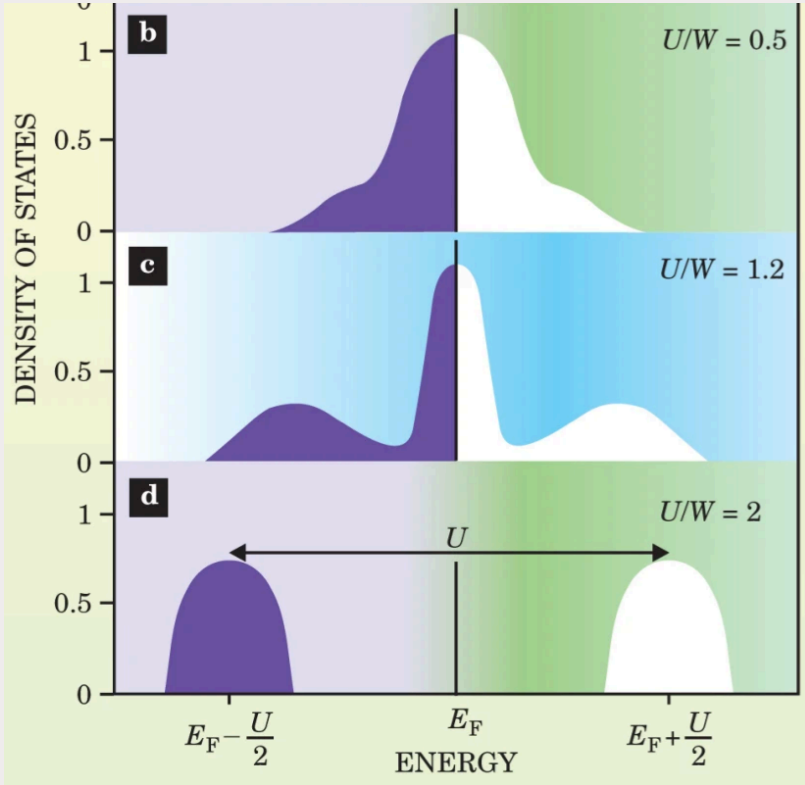

Phase Diagram and Impurity Phase Transition

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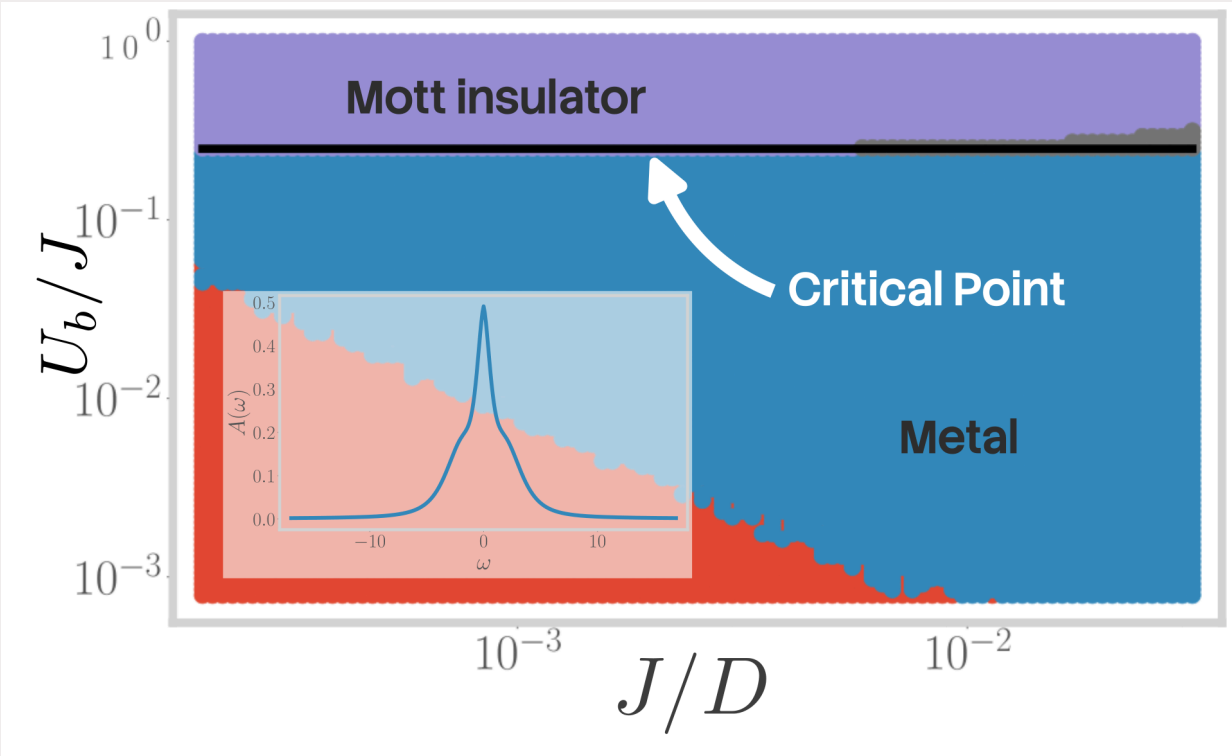

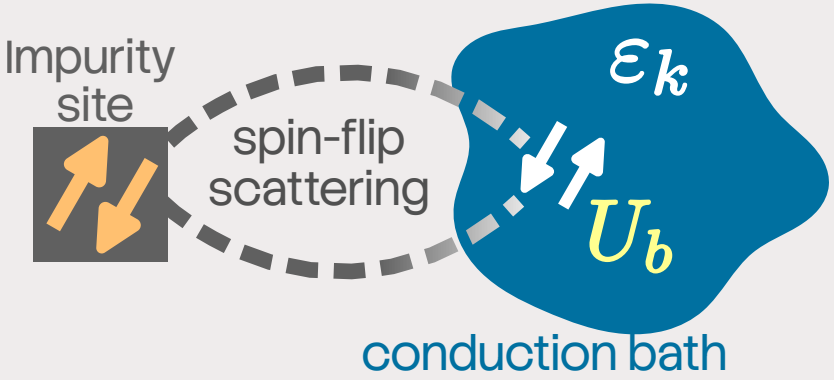

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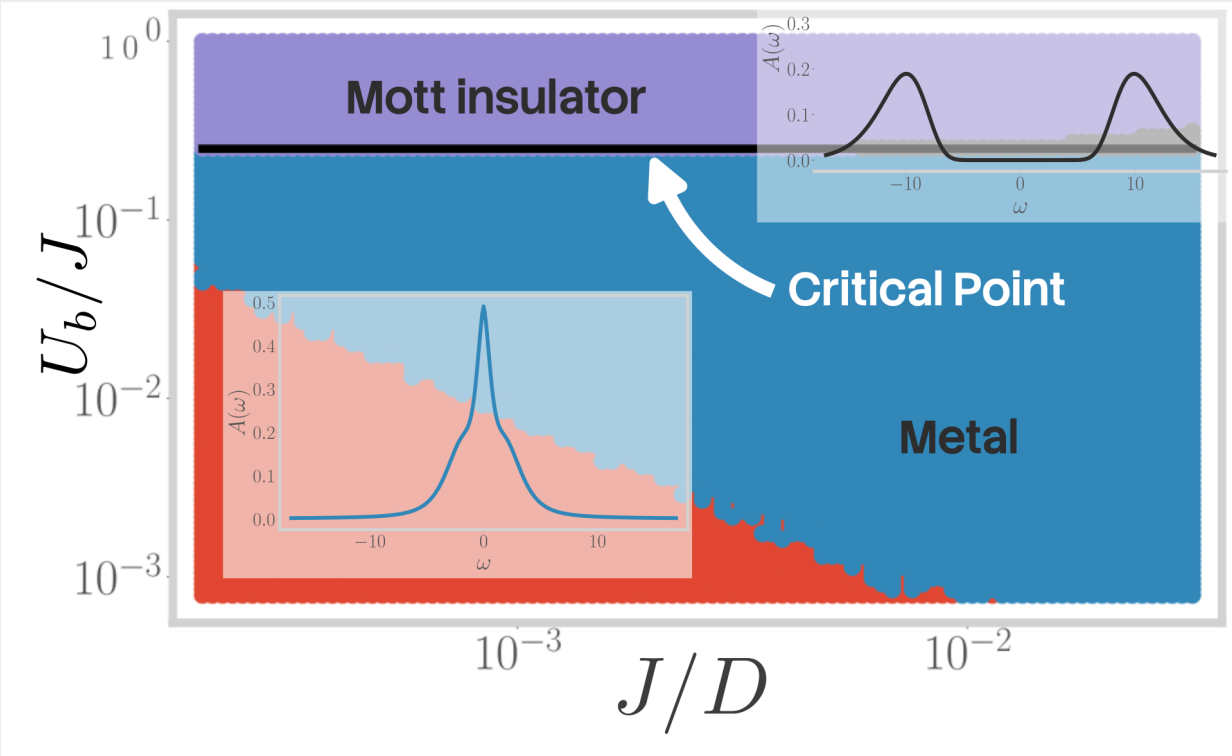

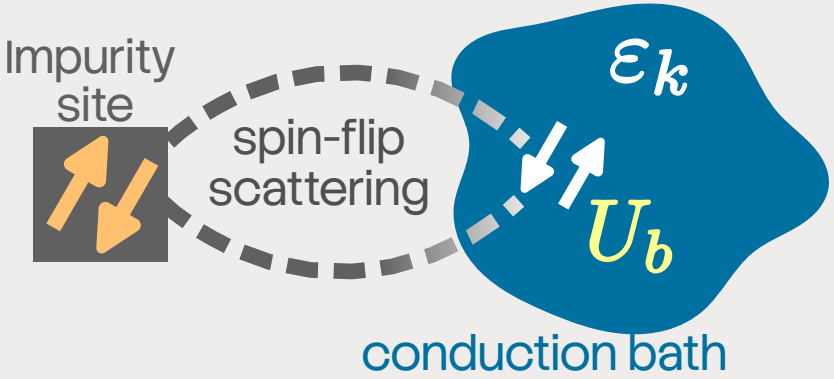

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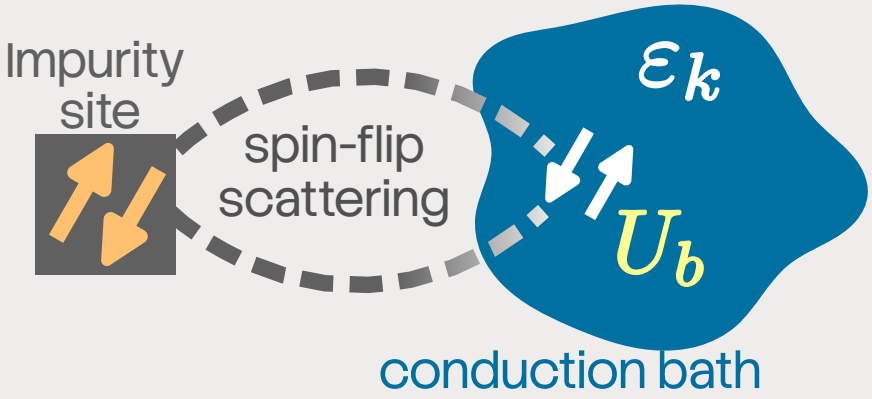
Local Attractive Bath Correlaton Is The Trick!

Local Attractive Bath Correlaton induces phase transition not present in vanilla impurity model.



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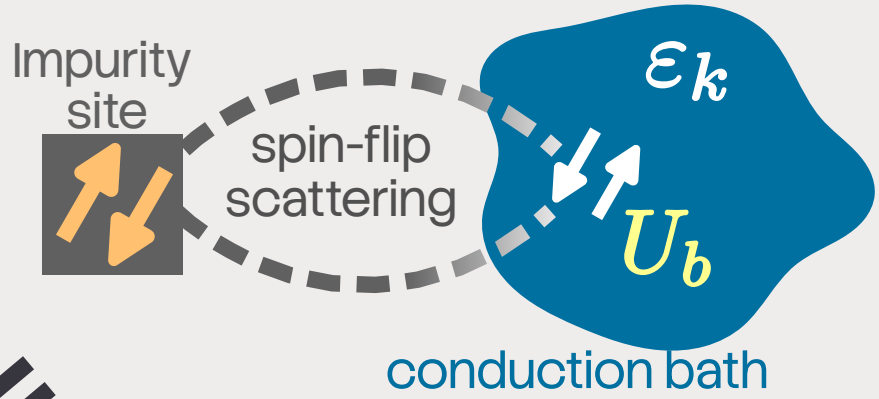

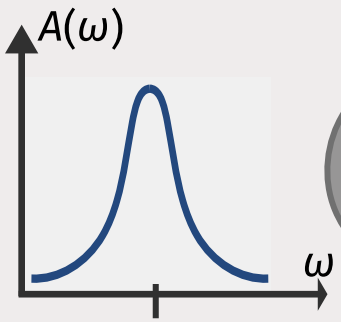

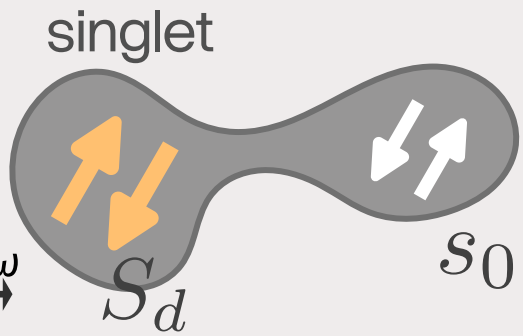

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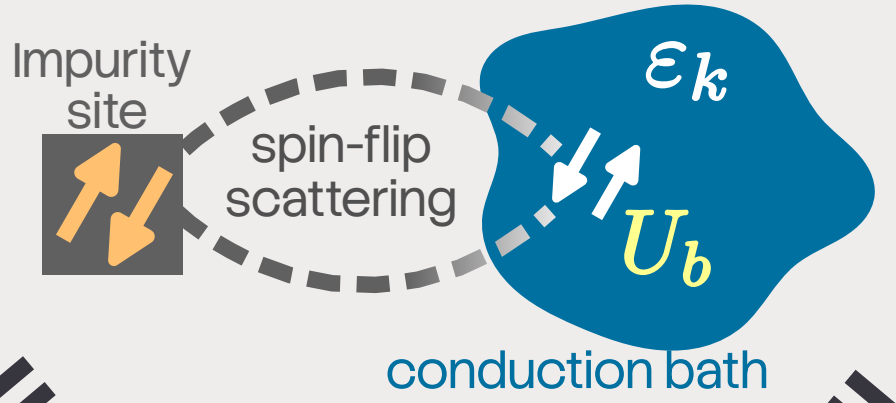

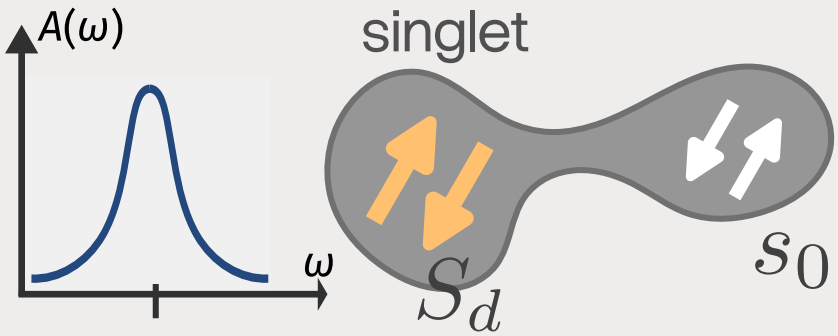

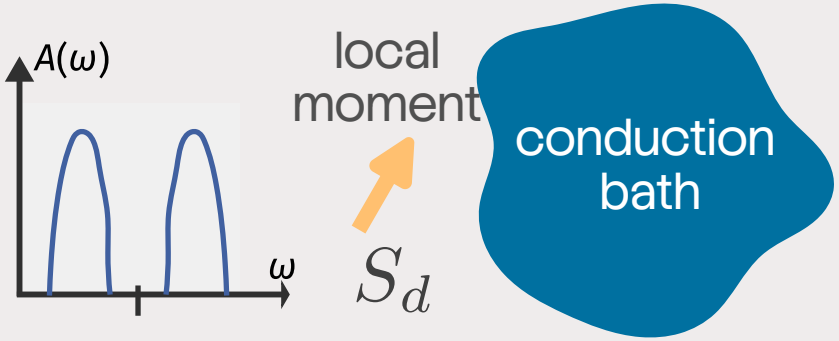

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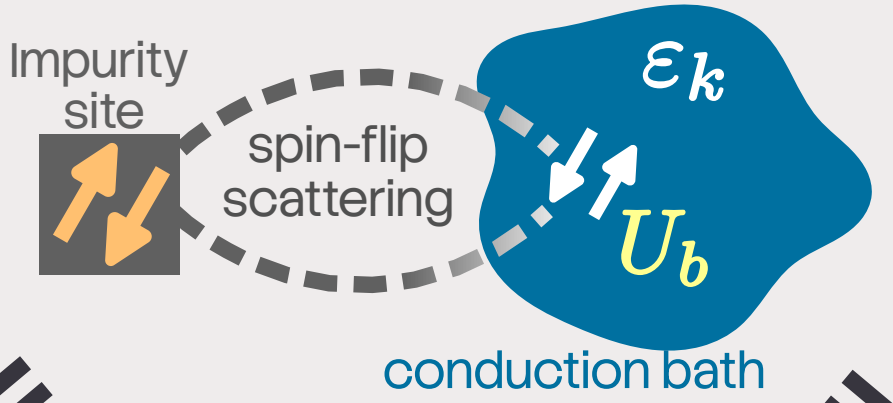
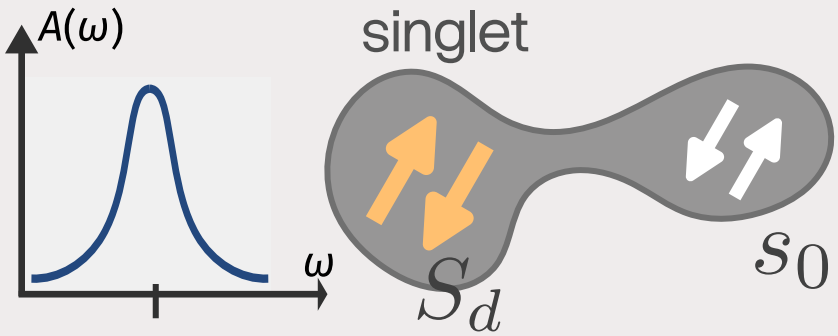
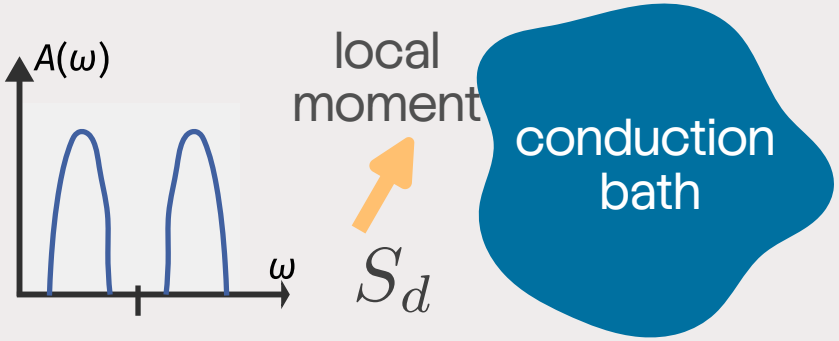


State exactly at transition is a **special metal** – more on this later!





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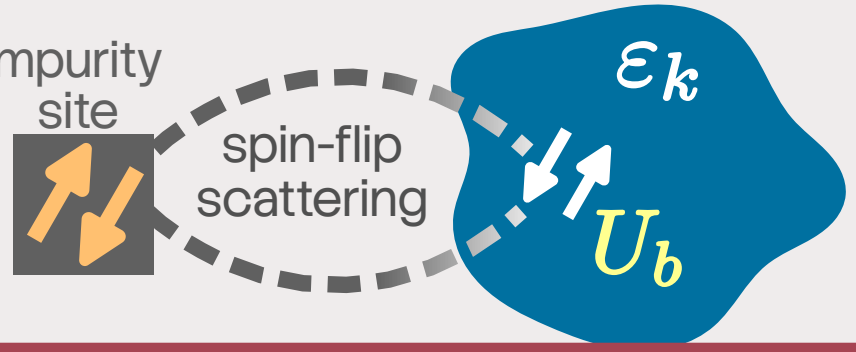

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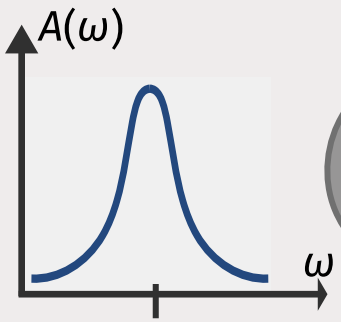
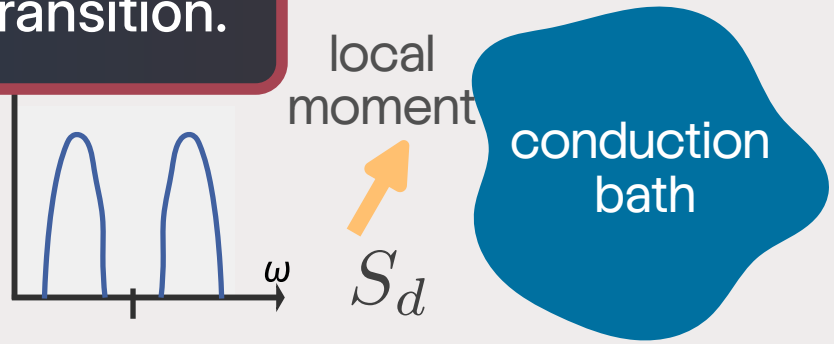
Impurity model with local attractive bath correlation is a good auxiliary system for Mott transition.



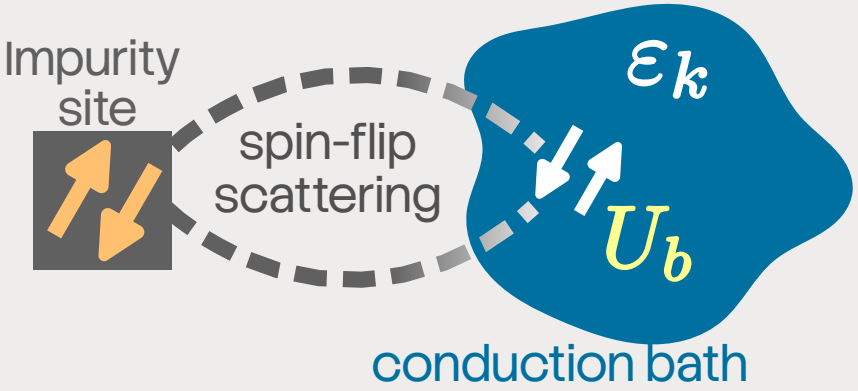


Capturing Spatial Fluctuations and Lattice Geometry

Impurity-Lattice Mapping Beyond Mean-Field

New J. Phys. 28 013503 (2026)
AM, S R. Hassan, A Mukherjee, N S. Vidhyadhiraja, A Taraphder, S Lal

Extending This To The More Interesting $D = 2$ Case



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- **non-local fluctuations vanish** in $d \longrightarrow \infty$ (mean-field is exact)
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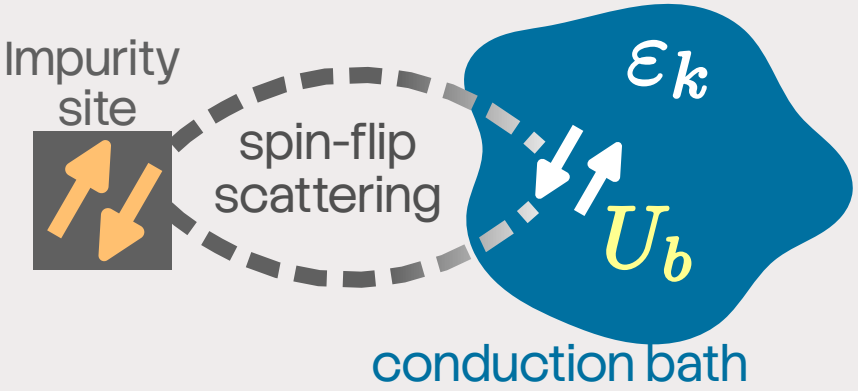

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Low-Dimensional Lattices Are More Interesting!

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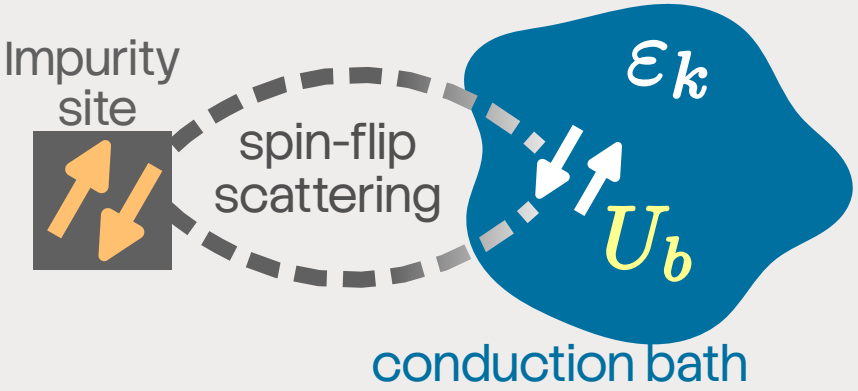

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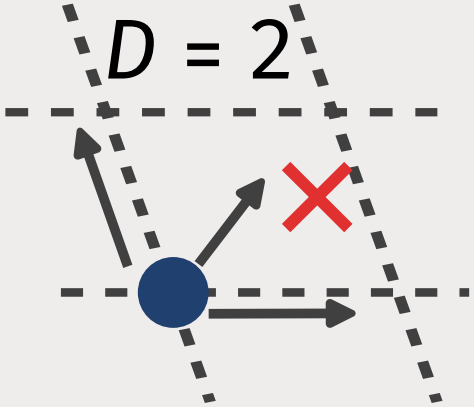

Paths extremely constrained

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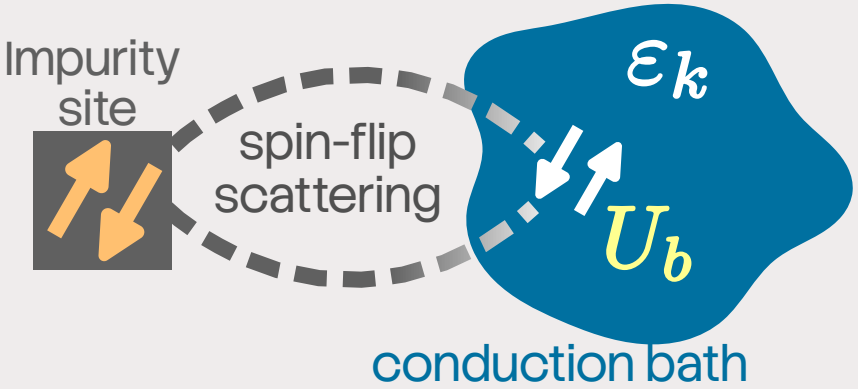

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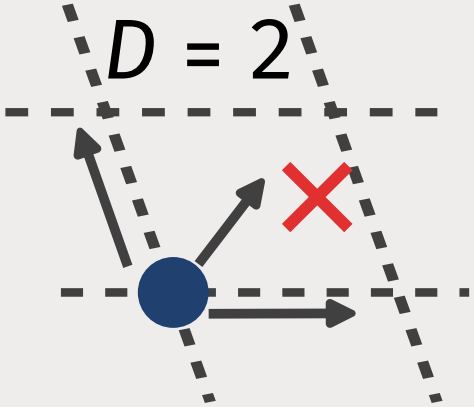
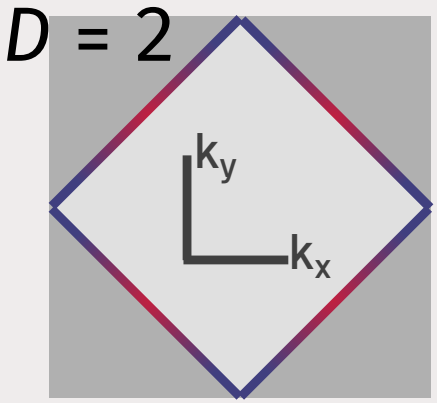
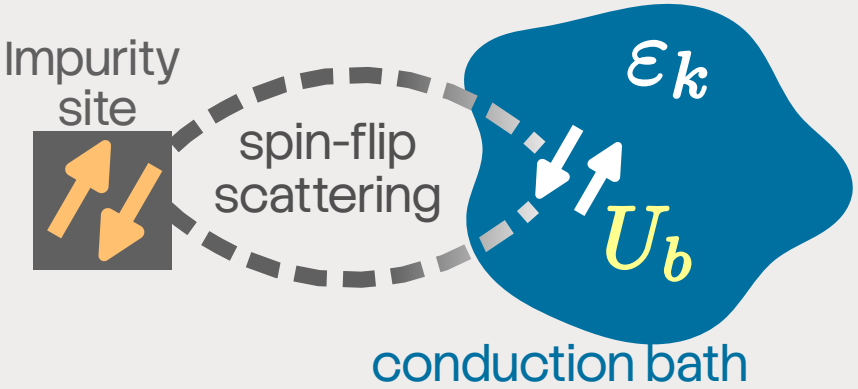


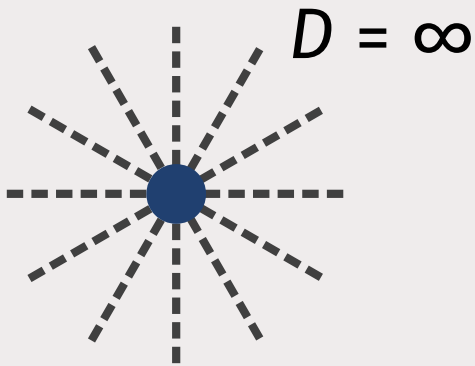
Paths extremely constrained



Momentum density-of-states highly anisotropic

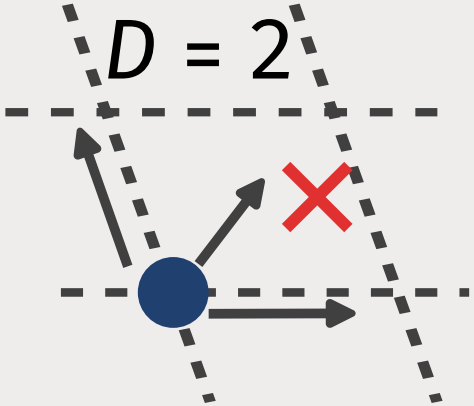
Extending This To The More Interesting $D = 2$ Case





- All paths symmetric
- No non-trivial k – dependence

Low-Dimensional Lattices Are More Interesting!



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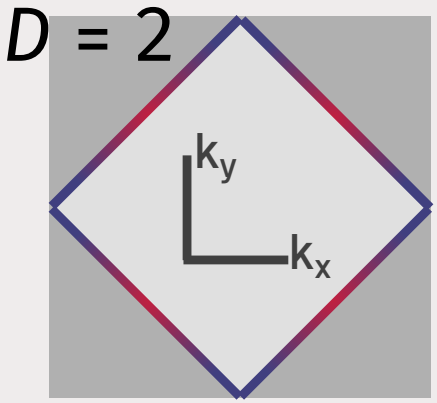

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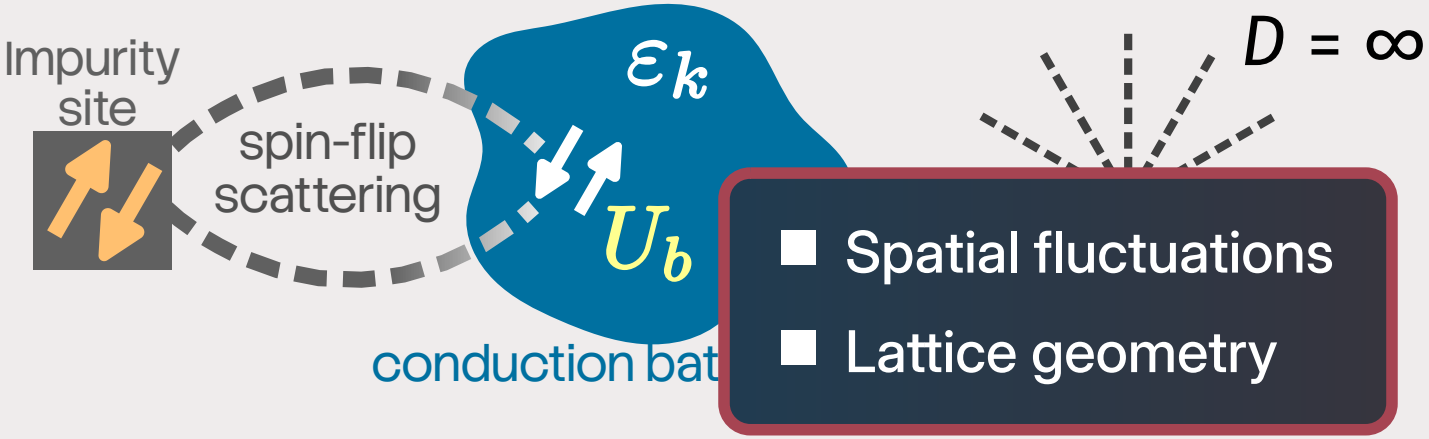

- Spatial fluctuations
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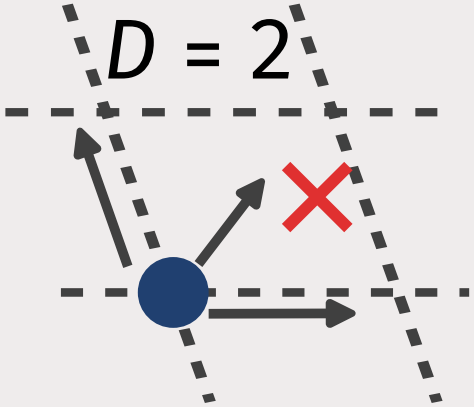

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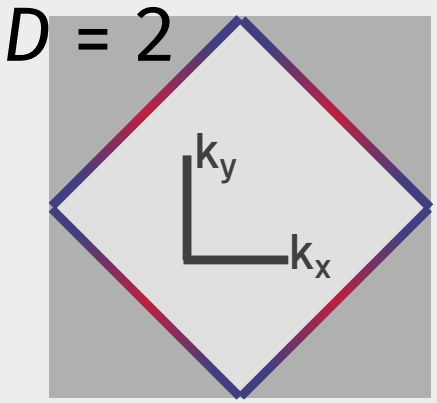

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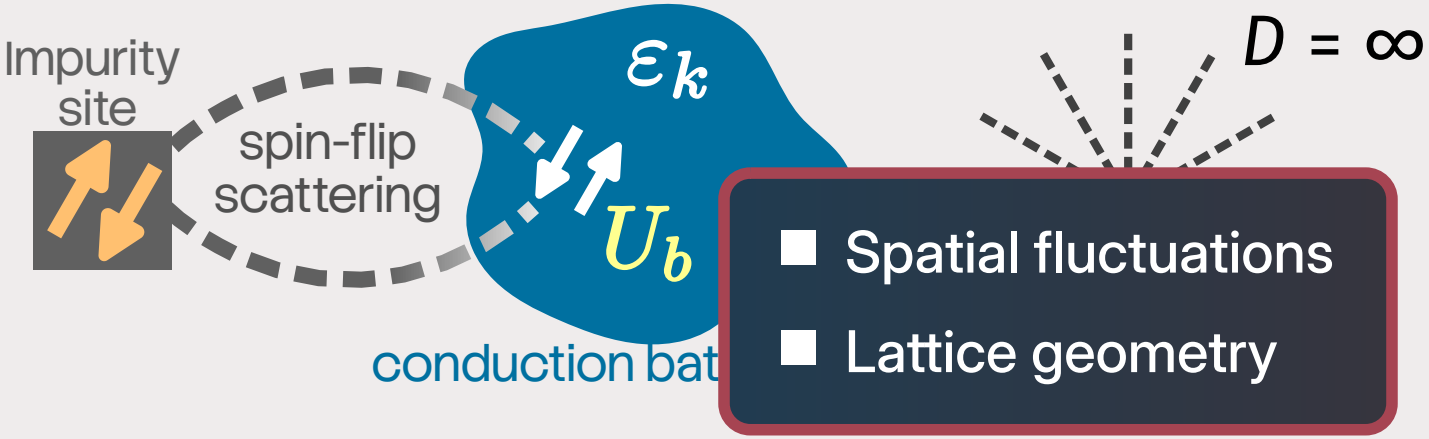

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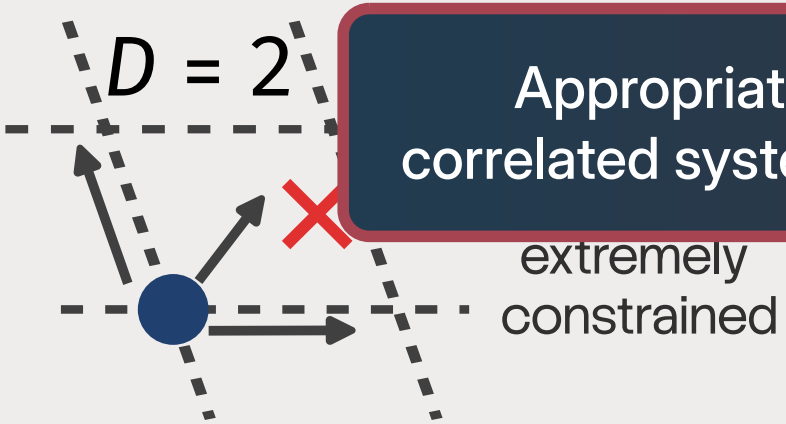
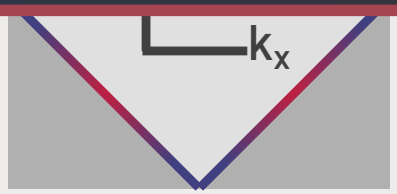


Appropriate auxiliary model for a correlated system on a 2D square lattice?



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Why is 2D Important?

- Many quantum materials are layered.
- Weak interlayer coupling $\longrightarrow$ effectively 2D

Canonical Example: Copper Oxide Materials –> Plethora of Mysterious Phases

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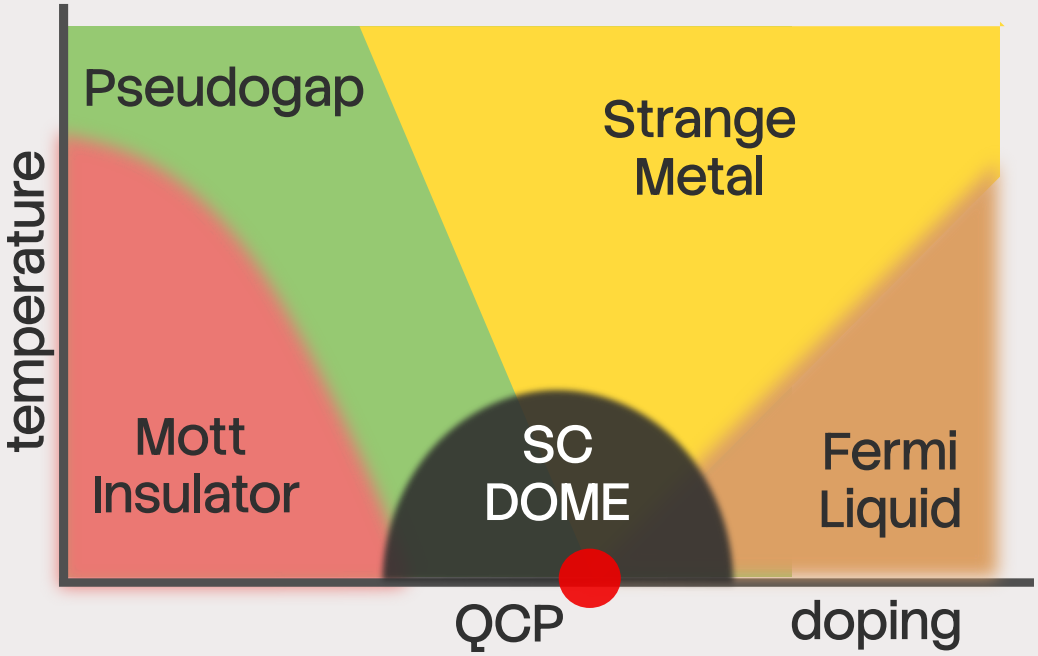

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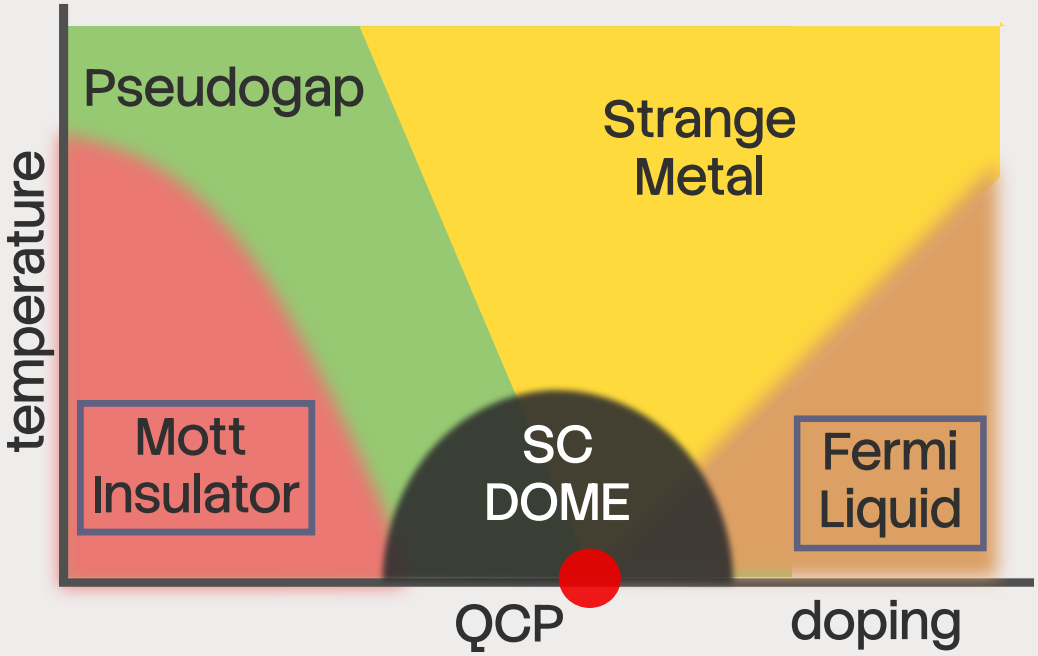

- Fermi liquid: conventional metal → described by **Landau Fermi-liquid theory**
- Excitations: long-lived **weakly-interacting** quasiparticles
- Mott insulator: interaction-driven insulator at half-filling (often AF ordered)
- Both endpoints are **well understood**

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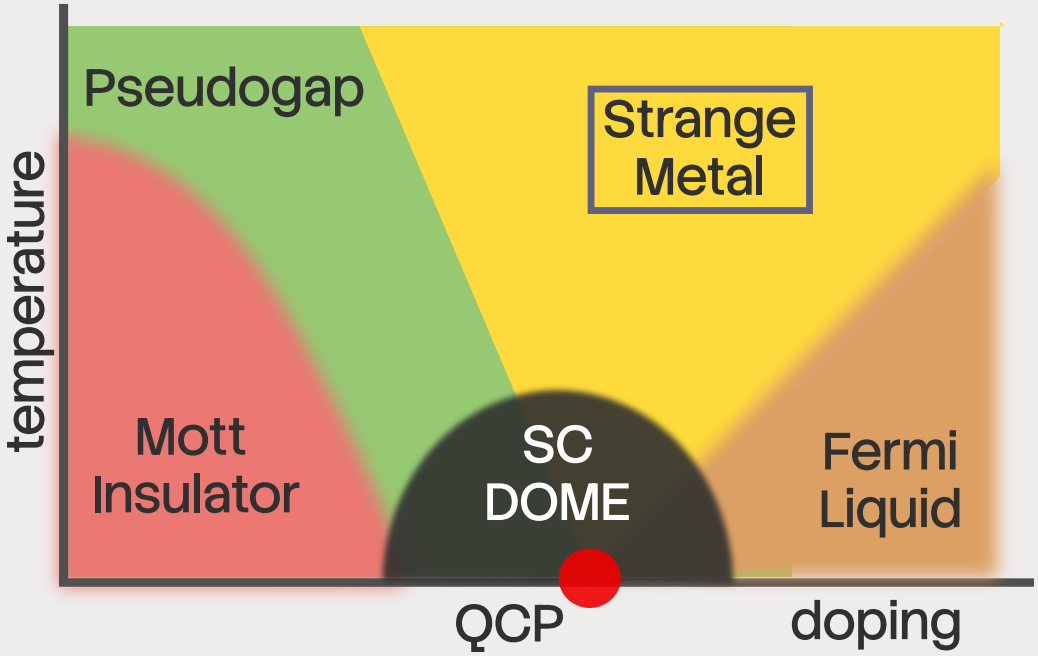

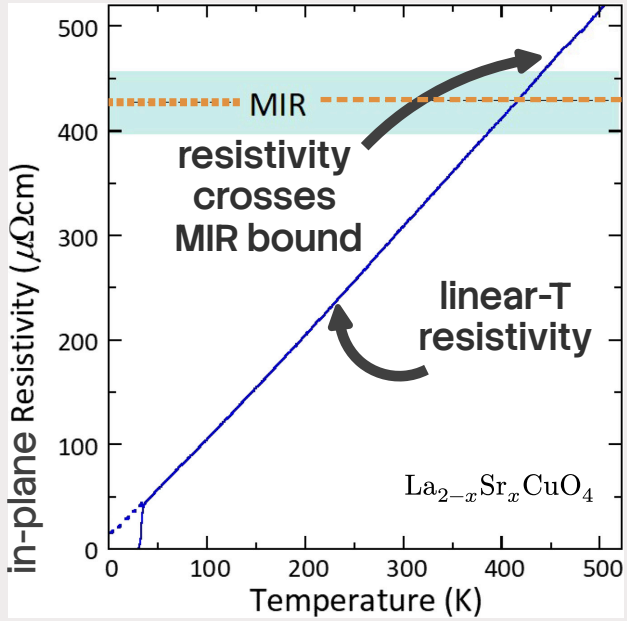

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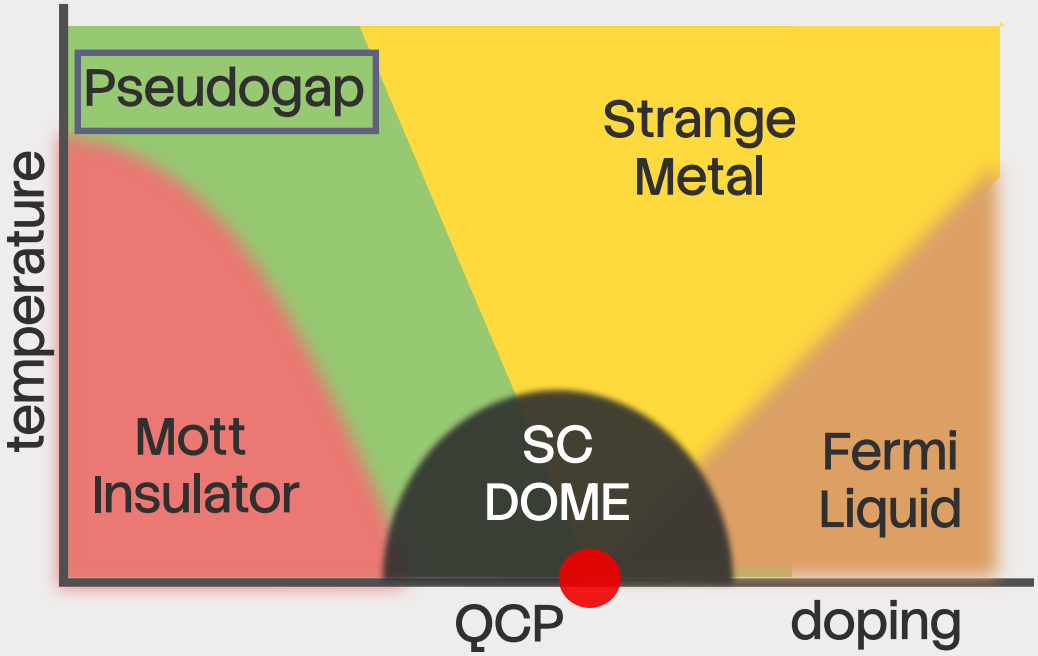
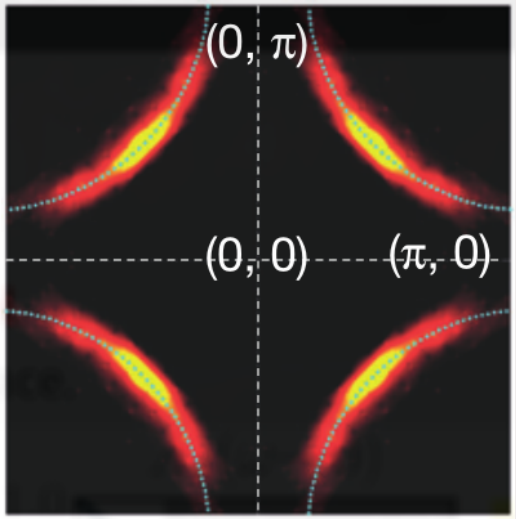


ARPES shows **partial gaps** on Fermi surface



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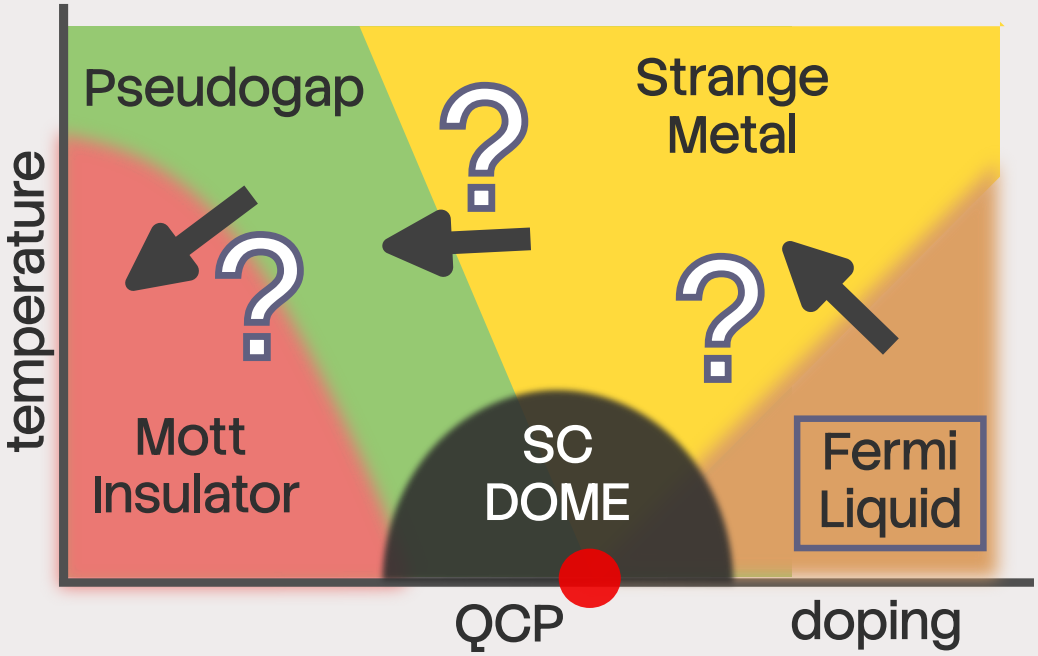
- How are the phases related to each other?

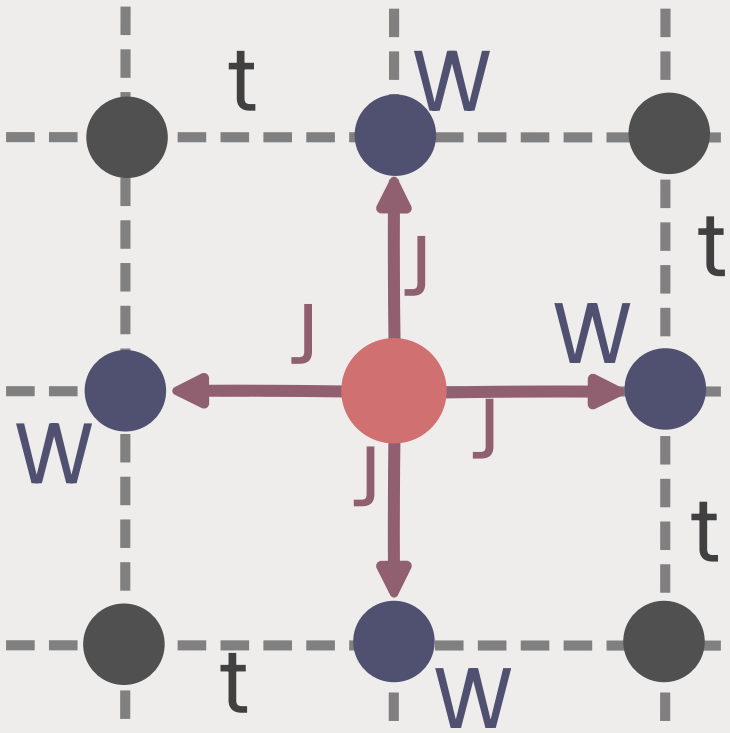


Inter-relations between the phases not clear!

- How do the electrons in Fermi liquid **morph** to form the strange meta/PG phases?
- How do the excitations of the PG phase **bind** to stabilise the Mott insulator?

Embedding An Impurity Into The Lattice

Embed an impurity into a 2D square lattice



- Impurity-Bath spin-interaction (J) can **probe lattice structure**

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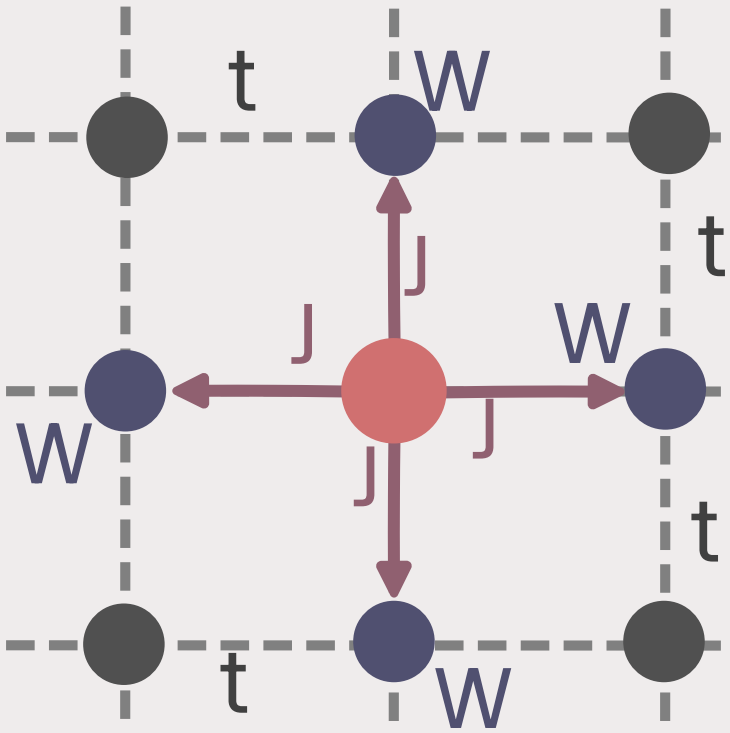
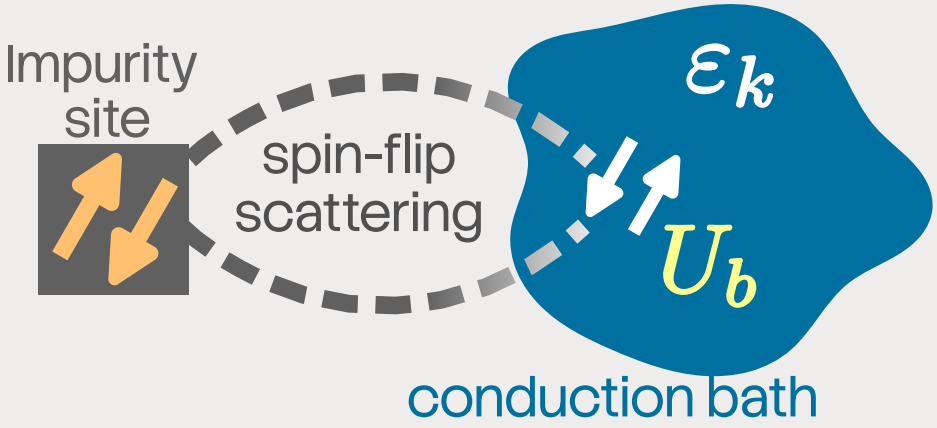


Contrast with $D = \infty$ limit



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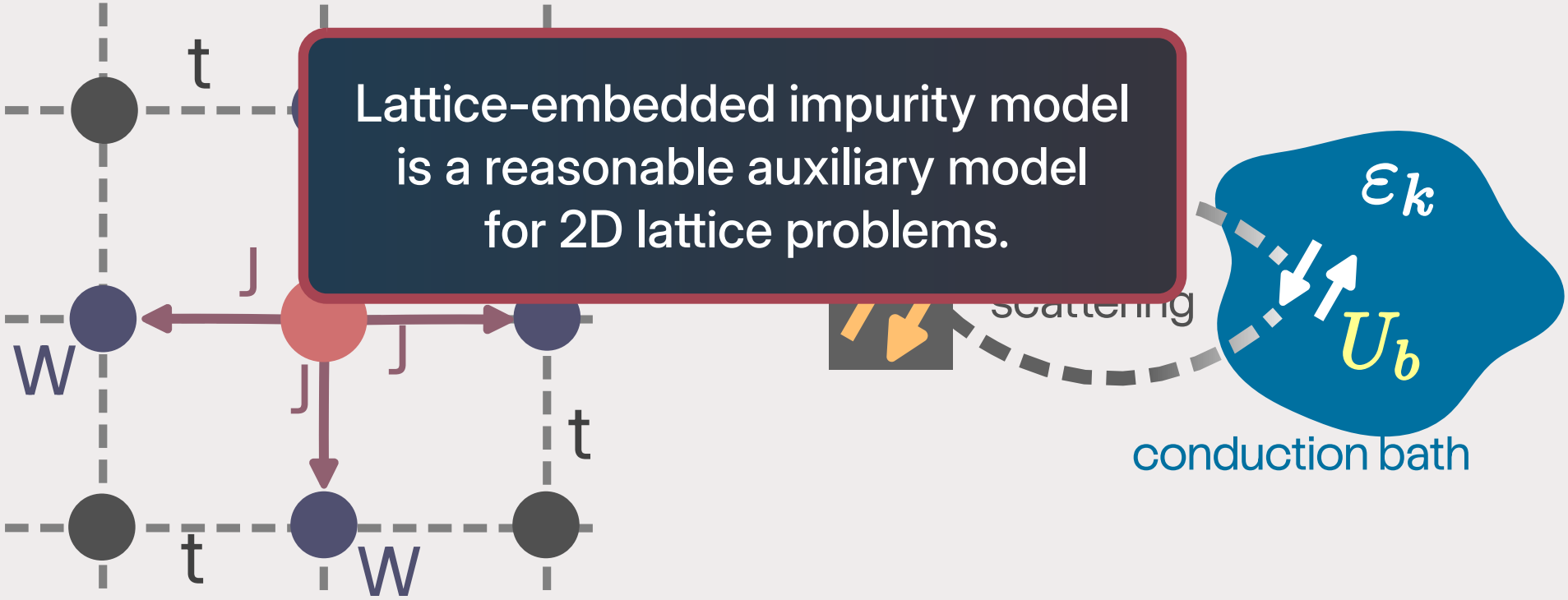

Lattice-embedded impurity model is a reasonable auxiliary model for 2D lattice problems.

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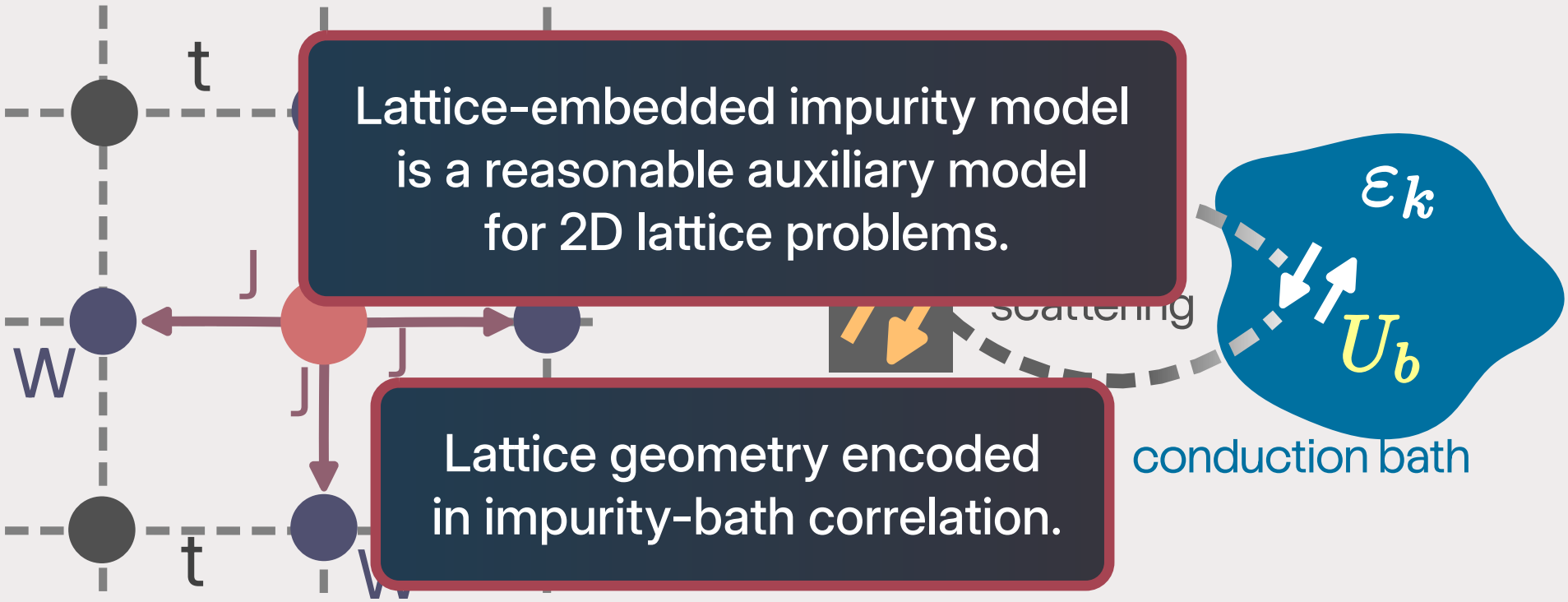

Lattice-embedded impurity model is a reasonable auxiliary model for 2D lattice problems.

Lattice geometry encoded in impurity-bath correlation.

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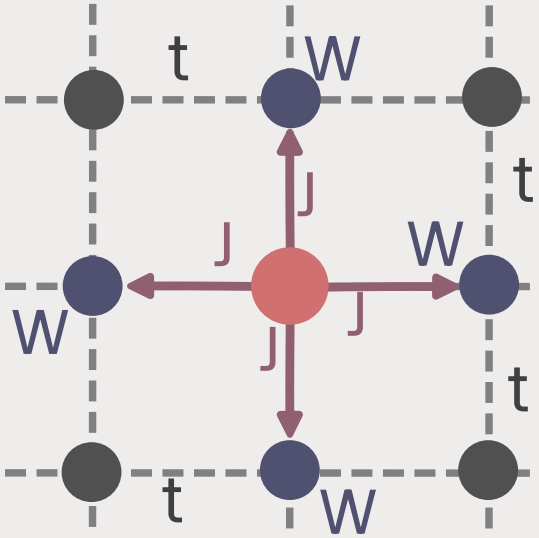
Reconstructing Spatial Fluctuations



$\Longrightarrow$

Lattice Translation symmetry needs to be restored

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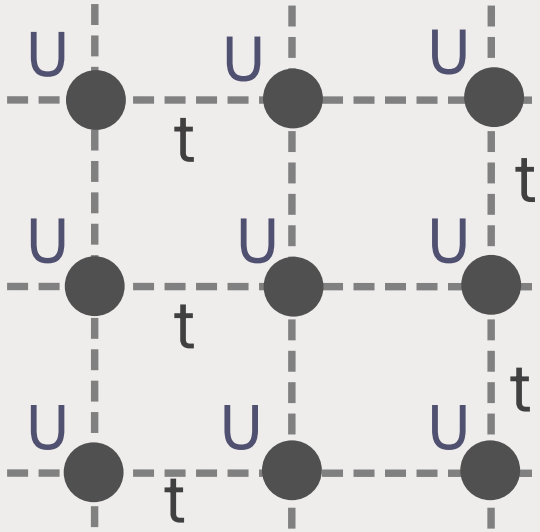

Reconstructing Spatial Fluctuations

Lattice Translation symmetry needs to be restored: <u>Bloch's Theorem!</u>

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Many-electron
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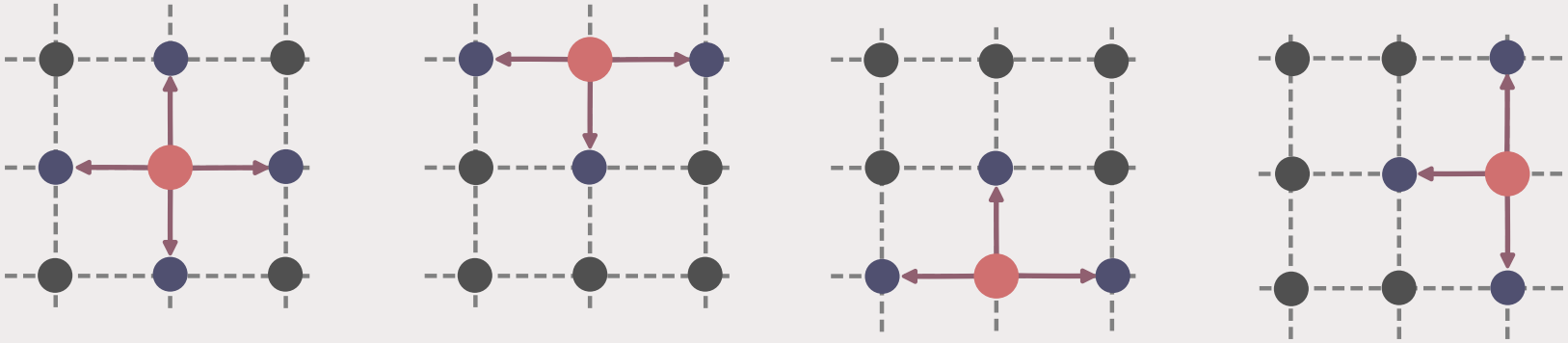
$\mathbb{T}(a) = \Pi_i T_i(a)$



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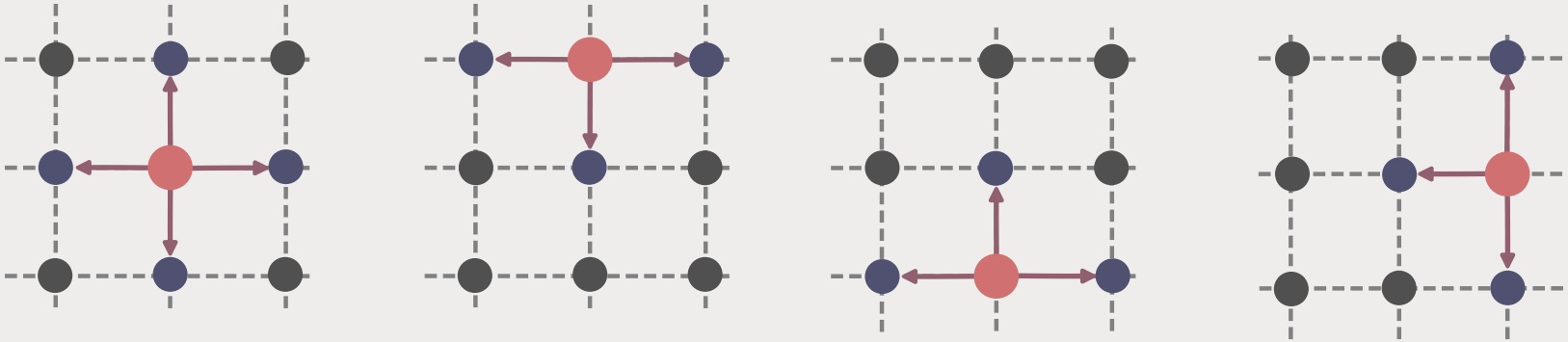

Many-electron translation:
Impurity → Lattice

$$H_{\text{lat}} = \sum_i \mathbb{T}(r_i) H_{\text{aux}} \mathbb{T}^\dagger(r_i)$$

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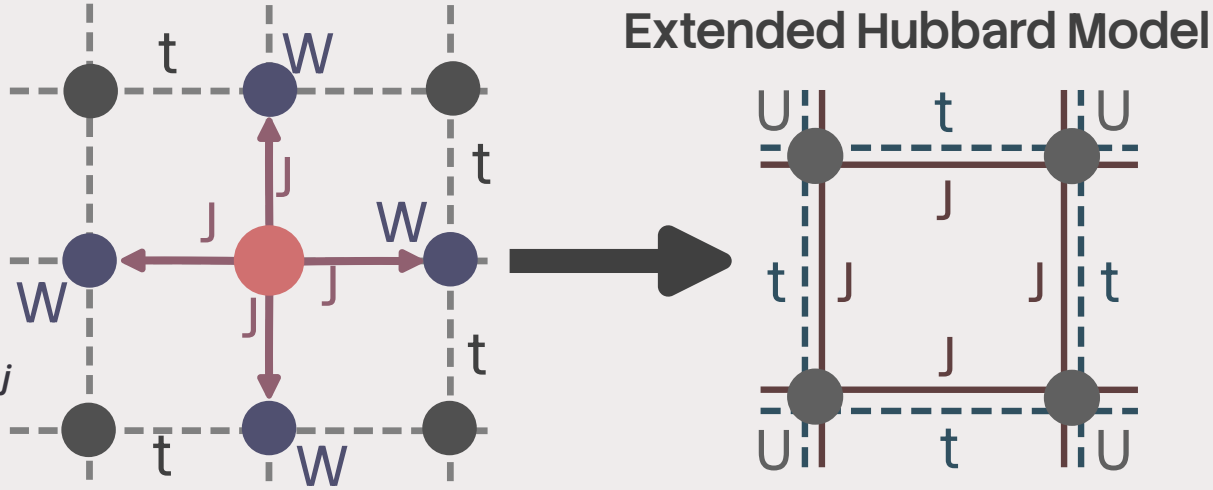

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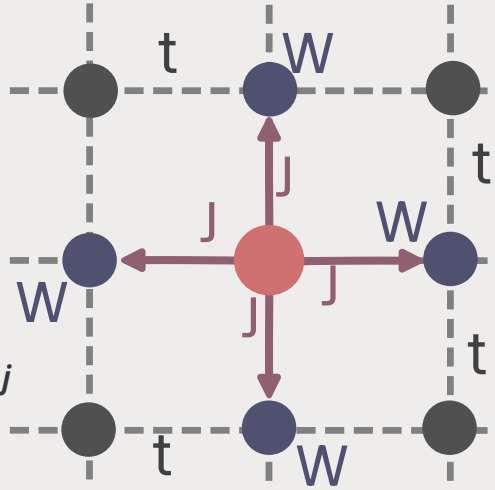
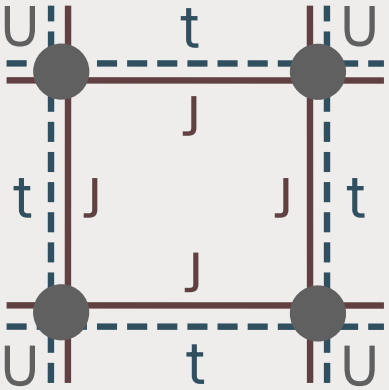


Extended Hubbard Model



Plan: Tiling Procedure + Many-body Bloch's theorem

Connects Hamiltonians, Eigenstates, Correlations

$$\Psi_{\text{lat}} = \sum_i \mathbb{T}(\mathbb{r}_i) \psi_{\text{aux}}$$

$$G_{\text{loc}} = \mathcal{G}(r_d, r_d) + C_0 \mathcal{G}(r_d, r_0) + C_1 \mathcal{G}(r_d, r_1) + \ldots$$

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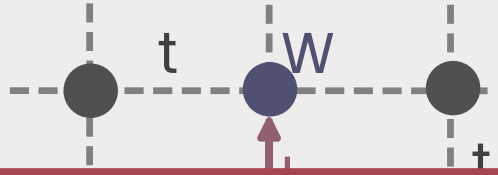

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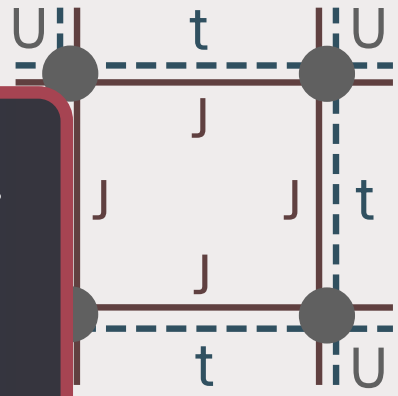

$$H_{\text{lat}} = \sum_i \mathbb{T}$$

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- Study lattice-embedded impurity model.
- Use tiling relations to reconstruct low-energy physics of lattice model.

Plan: Tiling Procedure + Many-body Bloch's theorem

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Phase Diagram: Lattice-Embedded Impurity Model

We use local spectral function to characterise phases

Competing tendencies In a Multi-Orbital System

Kondo screening vs. Mott localisation in a Heavy-Fermion model

(in progress)

AM, S Lal

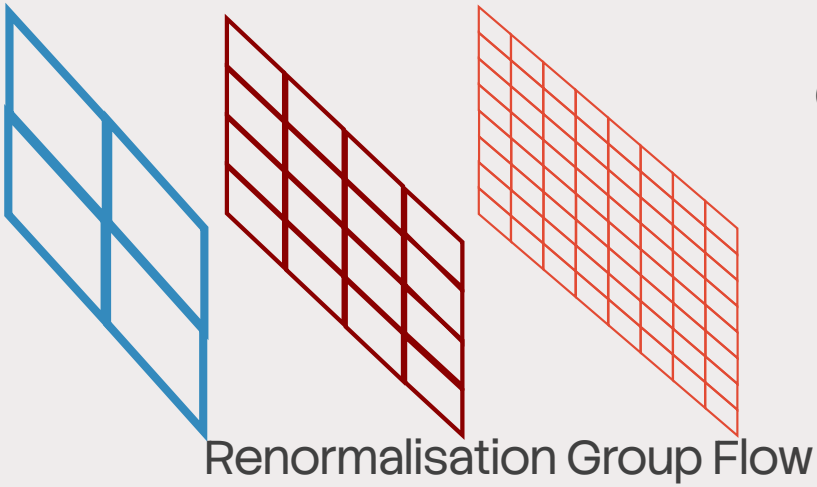
Holography of Entanglement in 2D Free Fermions

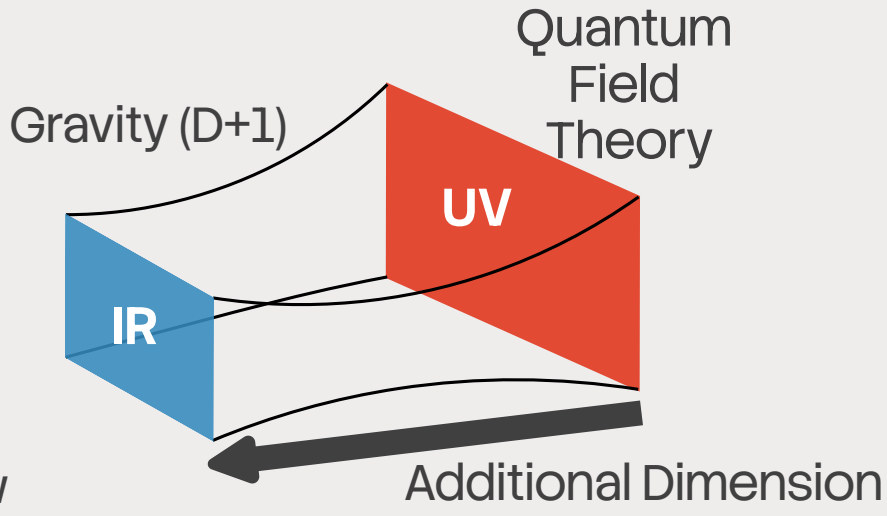
Emergent Geometry and Fermionic Criticality

J. Phys. A: Math. Theor. 57 275401 (2024)
AM, S Patra, S Lal

What is The Holographic Principle?

QUANTUM FIELD THEORY IN D – DIMENSIONS $\Longrightarrow$ GRAVITY THEORY IN $D+1$ – DIMENSIONS





- Extra dimension arises from **renormalisation group flow** of quantum theory
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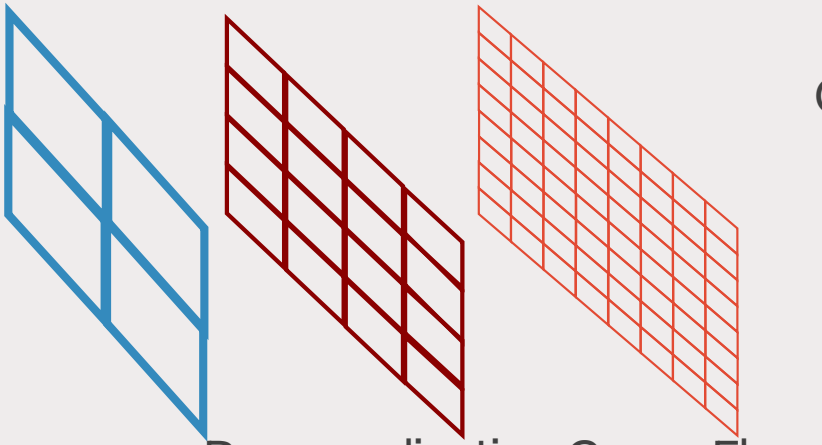
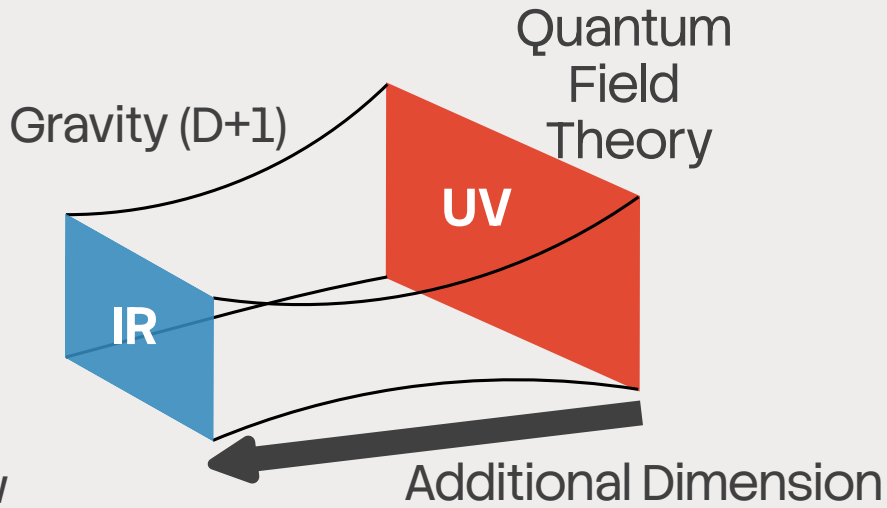
Quantum field theory in D – dimensions $\Longrightarrow$ Gravity theory in $D+1$ – dimensions



Renormalisation Group Flow



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Questions?

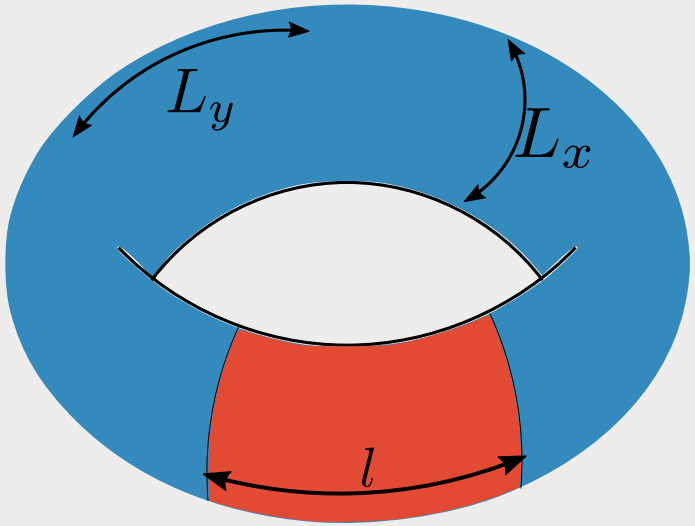
- Can we demonstrate this in the case of a prototypical metal-insulator transition?
- What effect does the critical point have on the emergent dimensions?

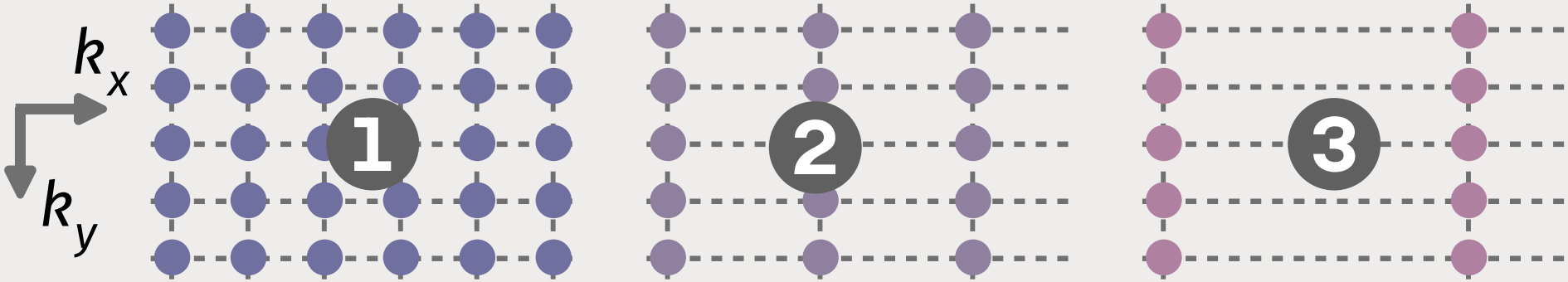
Entanglement RG Of 2D Dirac Electrons On a Torus

non-interacting **Dirac electrons** moving on a torus

- **Entanglement entropy** of a subsystem of length l
- RG: Apply **coarse-graining** transformation in k_x-space

Entanglement measure $S_{\mathrm{EE}}(j)$ at every step $\longrightarrow$ **RG Flow of entanglement**





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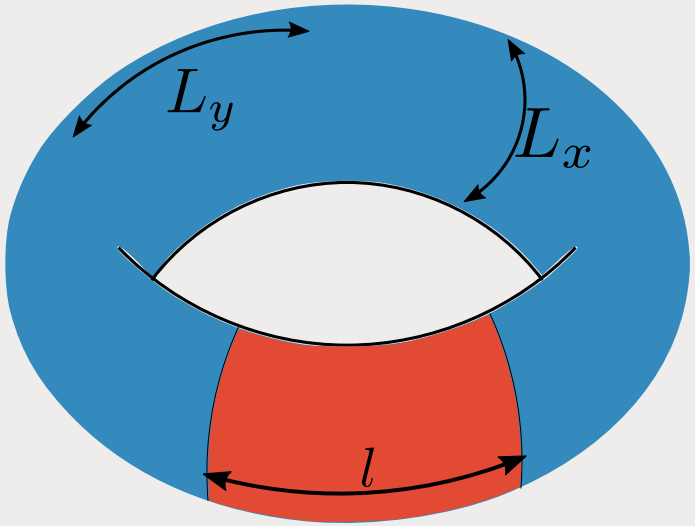
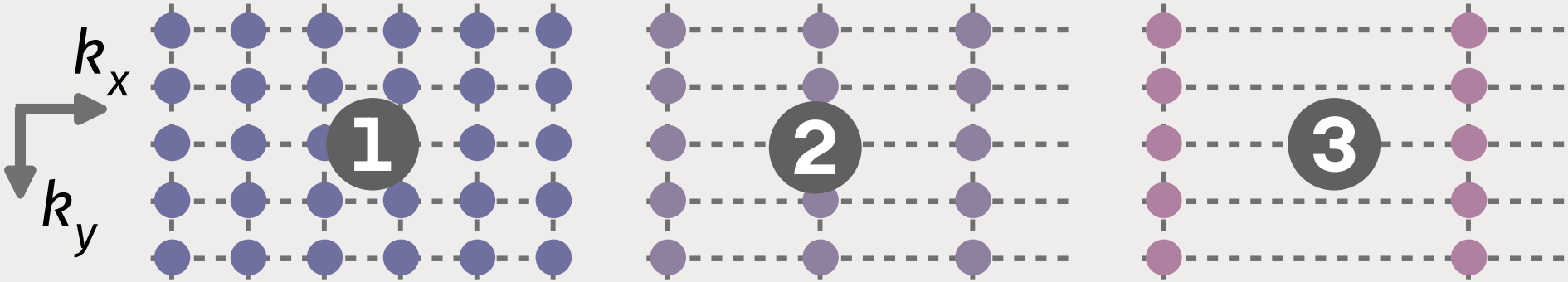
- RG: Apply **coarse-graining** transformation in k_x-space

Entanglement measure $S_{\mathrm{EE}}(j)$ at every step $\longrightarrow$ **RG Flow of entanglement**





Flow of Entanglement Defines Geometry

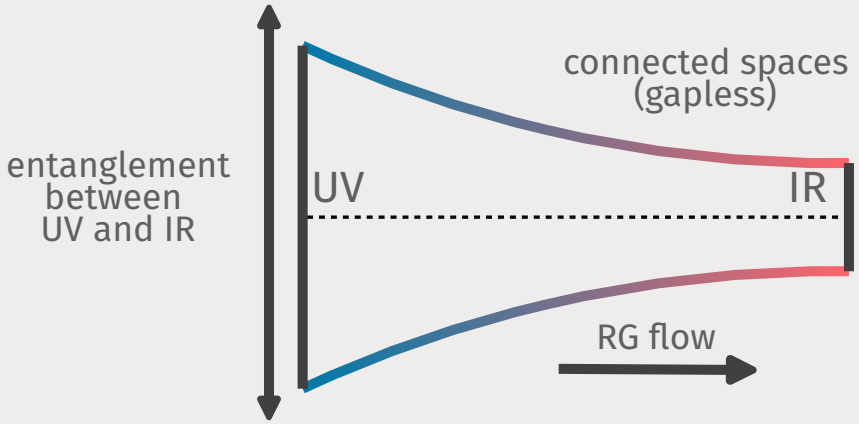
Define distance using **mutual information**:

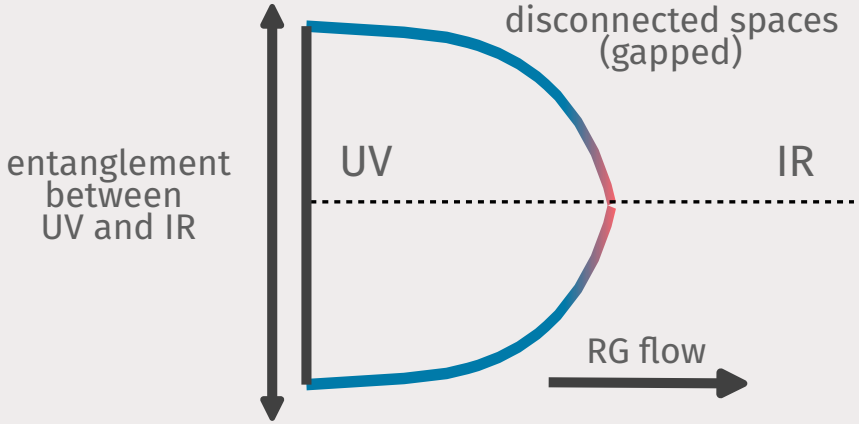
$$x_j = -\log(I_2(j+1) - I_2(j)) / \log(I_2(\max))$$

- high mutual information $\Longrightarrow$ small distance

Use distance to define curvature, metric, etc!

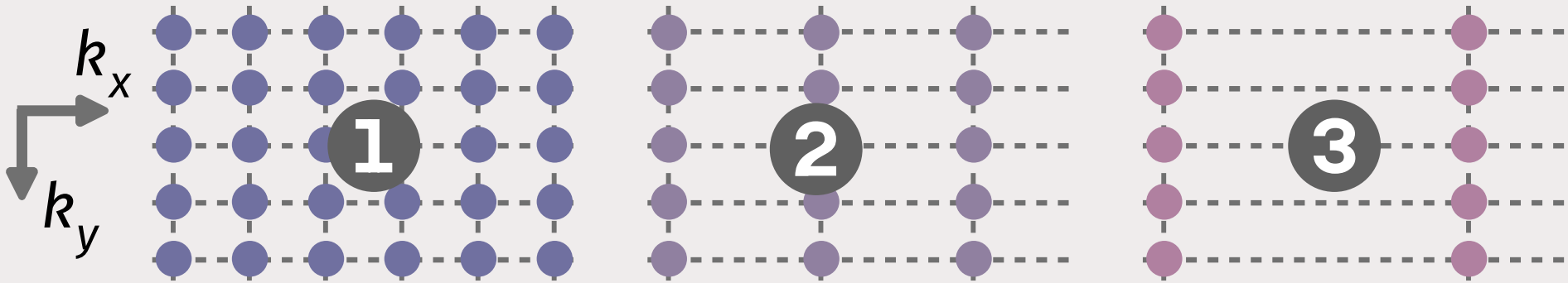
These define geometry of emergent extra dimension $\Longrightarrow$ can be used to obtain a **discrete metric**





What's happens at metal-insulator transition?

- geometry changes from open to closed
- wormhole-like geometry appears, between UV and IR


k_x
k_y
1
2
3

Closing Remarks

Summary of Primary Results

Main Conclusions and Takeaways

Future Work